Mobile and Autonomous Robots (UE23CS343BB7)

PERCEPTION

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Mobile and Autonomous Robots

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Perception in Autonomous Robots

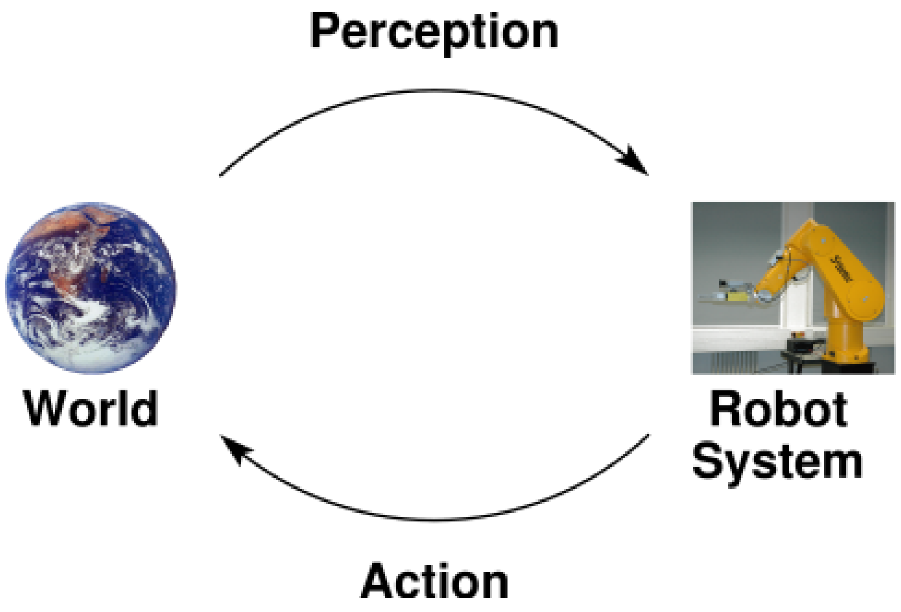
- Introduction to Perception
- Perception & Non-perception Sensors: Encoders, Gyroscope, Accelerometers, IMU, GPS

Perception in Robotics

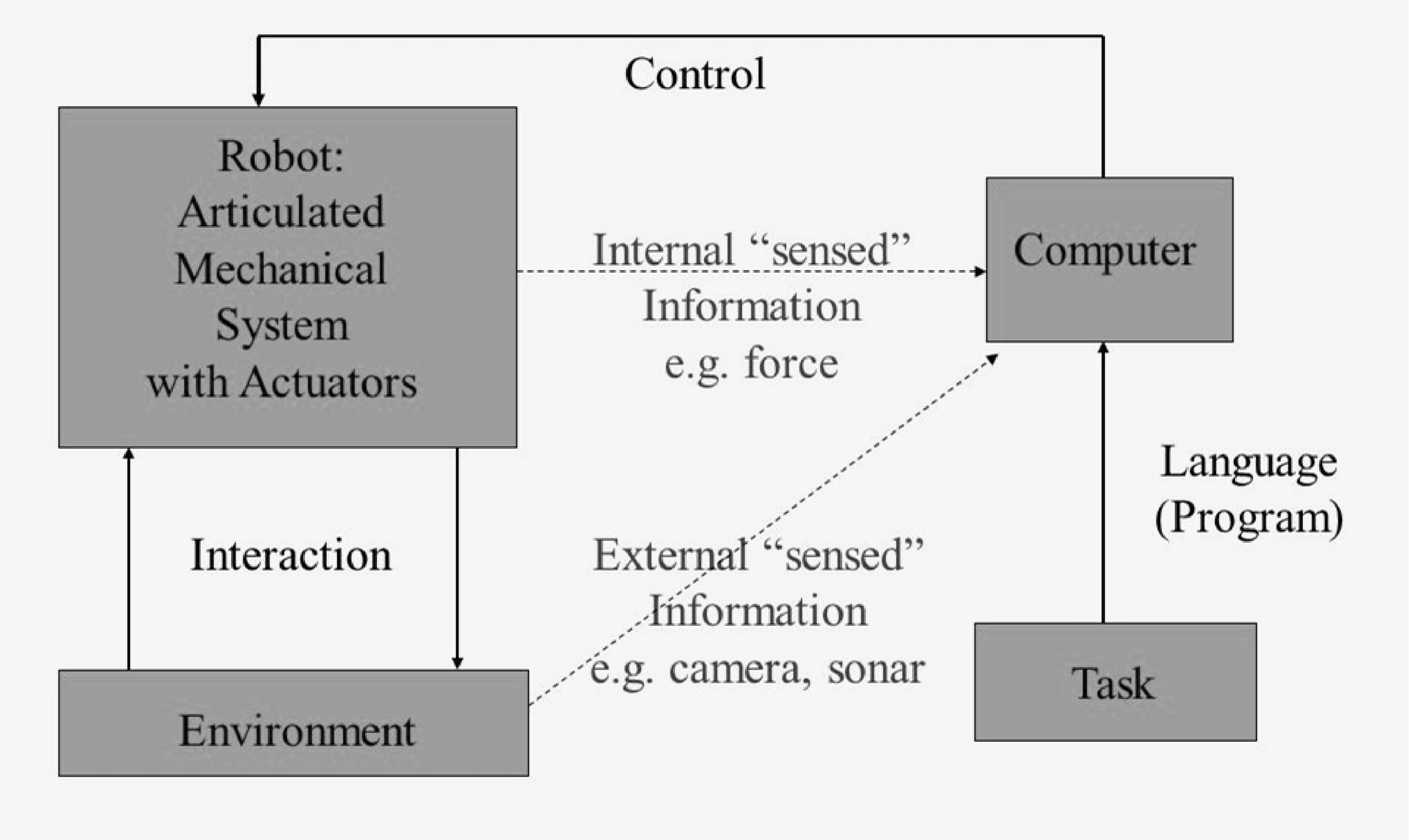
- Perception is the ability of a robot to sense, interpret, and understand its environment
- It converts raw sensor data into meaningful information
- Acts as the foundation for decision-making and autonomy

Perception enables a robot to answer the question: “What is happening around me?”

- One of the most important tasks of an autonomous system of any kind is to acquire knowledge about its environment.
- This is done by taking measurements using various sensors and then extracting meaningful information from those measurements, known as **Perception**.
- The capability of perception is entrusted to the sensory system, which can acquire data on the internal status of the mechanical system as well as the external status of the environment (force sensors, cameras, etc.).
- The ability to exert an action, both locomotion and manipulation, is provided by the actuation system, which animates the mechanical components of the robot.



- Components of a robotic system:
  - Sensory Inputs
  - Data Processing
  - Environment Understanding
  - Situation Assessment
  - Decision Making



Recap Sensors

- Sensors are devices that extract measurable quantities from surroundings.
- They can measure quantities internal or external to the sensor and is hence used for both internal feedback control and external interaction with the outside environment.
- Sensors can be used to acquire information about the robot's environment, internal temperature, rotational speed of motors or even to directly measure a robot's global position.
- Similarly, in a robot, as the links and joints move, sensors such as potentiometers and encoders send signals to the controller, allowing it to determine where each joint is located.
- Some of the types of sensors are:
  - Vision
  - Touch
  - Position
  - Range finders
  - Tactile
  - Proximity
  - Others like GPS, IMU and radioactive sensor

Sensor Classification

- We classify sensors using two important functional axes: ***proprioceptive/exteroceptive*** and ***passive/active*.**
- **Proprioceptive Sensors:** they measure values internal to the system (robot); for example, motor speed, wheel load, robot arm joint angles, battery voltage.
- Proprioception includes awareness of the robot's joint angles, its speeds, as well torques and forces.
- **Exteroceptive Sensors:** acquire information from the robot's environment; for example, measurements, light intensity, sound amplitude. Hence exteroceptive sensor measurements are interpreted by the robot in order to extract meaningful environmental features.
- Any form of sensation that results from stimuli located outside the body and is detected by exteroceptors, including vision, hearing, touch or pressure, heat, cold, pain, smell, and taste

- ***Passive* sensors** measure ambient environmental energy entering the sensor.
  - Examples of passive sensors include temperature probes, microphones, and CCD or CMOS cameras.
- ***Active* sensors** emit energy into the environment, then measure the environmental reaction. Because active sensors can manage more controlled interactions with the environment, they often achieve superior performance.
- However, active sensing introduces several risks: the outbound energy may affect the very characteristics that the sensor is attempting to measure. Furthermore, an active sensor may suffer from interference between its signal and those beyond its control.
- For example, signals emitted by other nearby robots, or similar sensors on the same robot, may influence the resulting measurements.
  - Examples of active sensors include wheel quadrature encoders, ultrasonic sensors, and laser range finders.

Sensors Characteristics

- Some sensors provide extreme accuracy in well-controlled laboratory settings, but are overcome with error when subjected to real-world environmental variations.

- **Cost**: must be balanced with other requirements of the design such as reliability, accuracy, repeatability, life and so on.

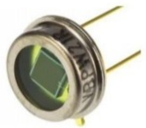
- **Weight**: Since robots are dynamic machines, the weight of a sensor is very important. A heavy sensor adds to the inertia of the arm and reduces its overall payload,
  e.g. a heavy camera mounted on a robotic insect would limit its flying capabilities.

- **Type of output (digital or analog):** Determines necessity of ADC and/or DAC.

Temperature Sensor

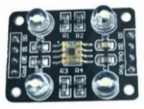
Humidity Sensor

Proximity Sensor



Light Sensor

Metal Sensor



Color Sensor

- **Resolution:** is the minimum difference between two values that can be detected by a sensor. Usually, the lower limit of the dynamic range of a sensor is equal to its resolution.

  - In a digital device with *n* bits, the resolution is:

$$\text{Resolution} = \frac{\text{Full Range}}{2^n}$$

Ex: What is the resolution of an absolute encoder with four bits?

four bits can report positions up to $2^4 = 16$ different levels.

Therefore, its resolution is $360^0 / 16 = 22.5^0$

- For example, suppose that you have a sensor that measures voltage, performs an analog-to-digital (A/D) conversion, and outputs the converted value as an 8-bit number linearly corresponding to between 0 and 5 V. If this sensor is truly linear, then it has $2^8 - 1$ outputs, and resolution of

  5V/255 =20mV

Sensors Characteristics

- **Range**: Difference between smallest to the largest outputs or inputs
  - Range of the output signal limited
  - Output signal will eventually reach a minimum or maximum on measurements reaching limits
- **Response time:** Time that sensor's output requires to reach a certain percentage of total change
  - Expressed as 95% of total change.
  - A special response time of 63.2% is called time constant.
  - Rise time is the time required between 10% and 90% of the final value.
  - Settling time is the time between 0% and 98% rise.

Range

- ***Dynamic range*** is used to measure the spread between the lower and upper limits of input values to the sensor while maintaining normal sensor operation.
- Formally, the dynamic range is the ratio of the maximum input value to the minimum measurable input value.
- Because this raw ratio can be unwieldy, it is usually measured in decibels, which are computed as ten times the common logarithm of the dynamic range.

1. Find the dynamic range of a current sensor that can measure as low as 1mA upto 20Amps.

- **Solution**: $10 \cdot \log\left[\frac{20}{0.001}\right] = 43 \text{ dB}$

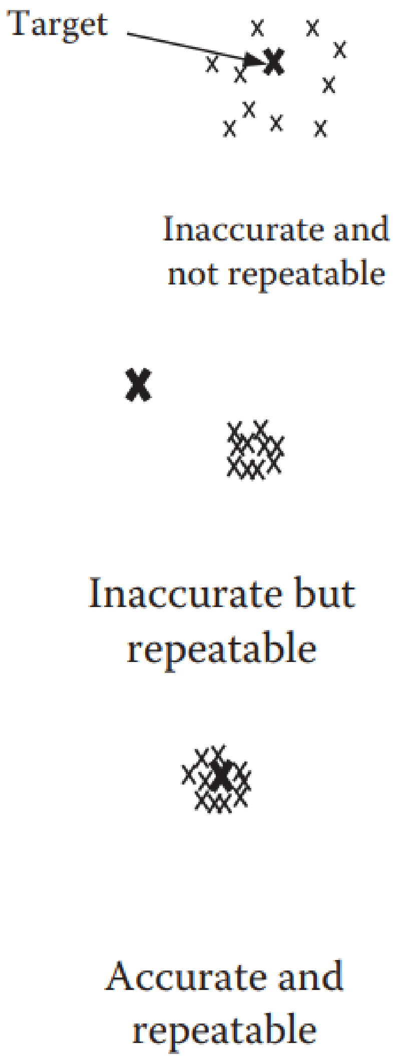
2. Find the dynamic range of a voltage sensor that can measure as low as 1mV upto 20V.

- **Solution:** Voltage is not a unit of power, but the square of voltage is proportional to power. Therefore, we use 20 instead of 10

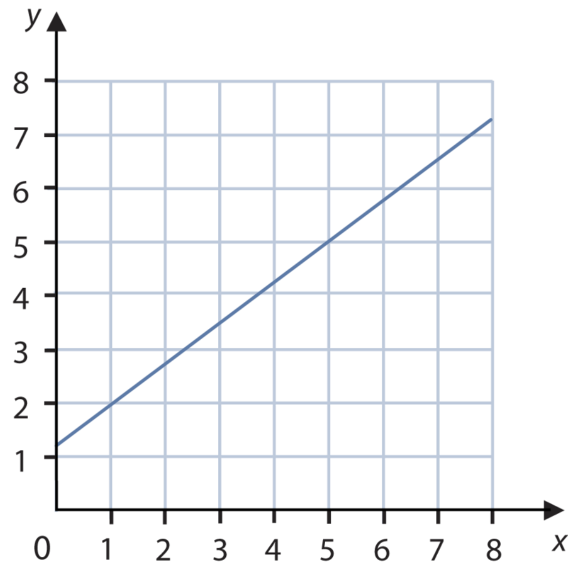
$$20 \cdot \log\left[\frac{20}{0.001}\right] = 86 \text{ dB}$$

- **Frequency response:**
  - All systems can resonate at around their natural frequency with little effort.
  - The frequency response is the range in which the system's ability to resonate (respond) to the input remains relatively high.
  - As the input frequency deviates from the natural frequency of system, the response falls off.
  - The larger the range of the frequency response, the better the ability of the system to respond to varying input.
  - Otherwise, the quickly changing variations in the phenomenon may not be measured by the sensor.
  - Therefore, it is important to consider the frequency response of a sensor and determine whether or not the sensor's response is fast enough under all operating conditions

- **Reliability:**
  - The ratio of how many times the system operates properly divided by how many times it is used.
  - For continuous and satisfactory operation, it is necessary to choose reliable sensors that last a long time while considering cost and other requirements
- **Accuracy:**
  - Defined as how close the output of a sensor is compared to the expected value
- **Repeatability (precision):**
  - Output of a sensor to similar input values may vary.
  - Defined as the radius of a circle that encompasses all output values if sufficient measurements are made



- **Linearity:** the relationship between input variations and output variations.
  - In a sensor with linear output, the same change in input at any level within the range will produce a similar change in output
    - (Almost all devices in nature are somewhat nonlinear, with varying degrees of nonlinearity)



- A linear response indicates that if two inputs *x* and *y* result in the two outputs f(x) and f(y), then $f(ax + by) = af(x) + bf(y)$.

- ***Bandwidth* or *frequency* is** used to measure the speed with which a sensor can provide a stream of readings. Formally, the number of measurements per second is defined as the sensor's frequency in *hertz.*
- Because of the dynamics of moving through their environment, mobile robots often are limited in maximum speed by the bandwidth of their obstacle detection sensors.
- Thus, increasing the bandwidth of ranging and vision-based sensors has been a high-priority goal in the robotics community

- **Sensitivity:** is a measure of the degree to which an incremental change in the target input signal changes the output signal. Formally, sensitivity is the ratio of output change to input change.
- Highly sensitive sensors will show larger fluctuations in output as a result of fluctuations in input.

- **Cross-sensitivity** is the technical term for sensitivity to environmental parameters that are orthogonal to the target parameters for the sensor.
- For example, a flux-gate compass can demonstrate high sensitivity to magnetic north and is therefore of use for mobile robot navigation. However, the compass will also demonstrate high sensitivity to ferrous building materials, so much so that its cross-sensitivity often makes the sensor useless in some indoor environments.

- **Error:** of a sensor is defined as the difference between the sensor's output measurements and the true values being measured, within some specific operating context.
- Given a true value *v* and a measured value *m*, error= *m – v .*

$$\left(accuracy = 1 - \frac{|error|}{v} \right)$$

- **Types of Errors:**
    - **Systematic errors:** are caused by factors or processes that can in theory be modeled. These errors are, therefore, deterministic (i.e., predictable). Poor calibration of a laser range, a bent stereo camera head due to an earlier collision are all possible causes of systematic sensor errors.

    - **Random errors:** cannot be predicted using a sophisticated model nor can they be mitigated by more precise sensor machinery. These errors can only be described in probabilistic terms (i.e., stochastically). Hue instability in a colour camera, spurious range finding errors, and black level noise in a camera are all examples of random errors.

- **Precision:** high precision relates to reproducibility of the sensor results.
- For example, one sensor taking multiple readings of the same environmental state has high precision if it produces the same output. In another example, multiple copies of this sensor taking readings of the same environmental state have high precision if their outputs agree.
- Precision does not, however, have any bearing on the accuracy of the sensor's output with respect to the true value being measured.
- Suppose that the *random error* of a sensor is characterized by mean μ and standard deviation σ, then:

$$precision = \frac{range}{\sigma}$$

Non-Perception Sensors

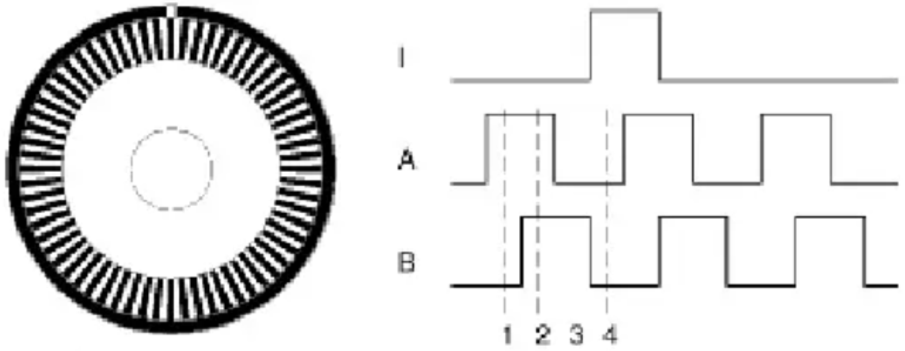
- Encoders
- Gyroscope
- Accelerometer
- IMU (Inertial Measurement Unit)
- GPS / GNSS / RTK-GPS

Perception Sensors

- RGB / Stereo / Depth Cameras
- LiDAR (2D / 3D)
- RADAR
- Ultrasonic Sensors
- Infrared / Thermal Cameras

Optical Encoders

- **Optical Encoders** are used to measure angular speed and position.
- These sensors are proprioceptive, and their estimate of position is done using wheel motion and best in the reference frame of the robot- **Odometry.**
- An optical encoder is basically a mechanical light chopper that produces a certain number of sine or square wave pulses for each shaft revolution.
- It consists of an illumination source, a fixed grating that masks the light, a rotor disc with a fine optical grid that rotates with the shaft, and fixed optical detectors.
- As the rotor moves, the amount of light striking the optical detectors varies based on the alignment of the fixed and moving gratings.

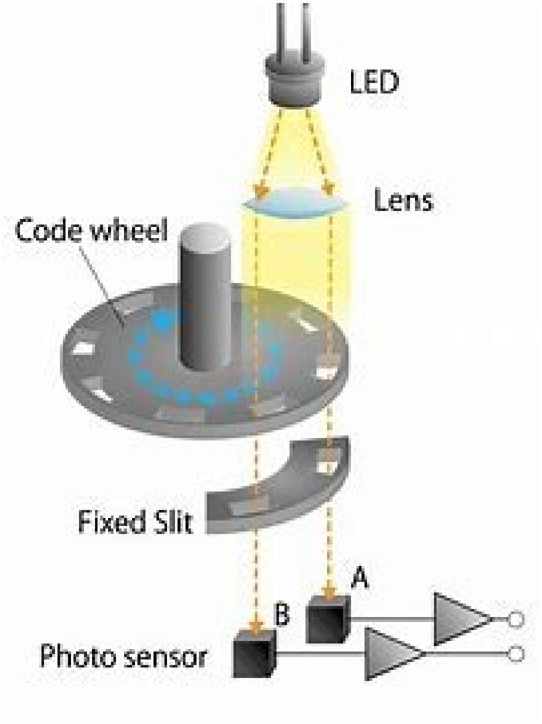


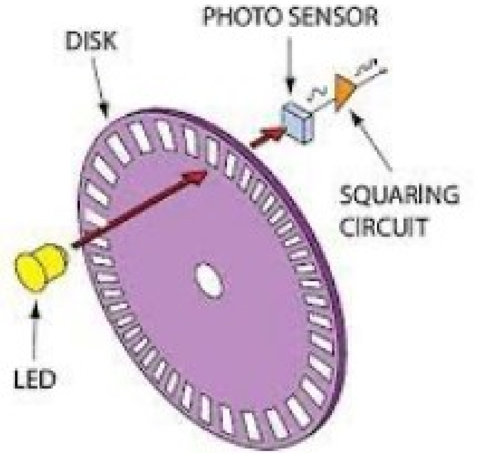
State	*Ch A*	*Ch B*
s_1	*high*	*low*
s_2	*high*	*high*
s_3	*low*	*high*
s_4	*low*	*low*

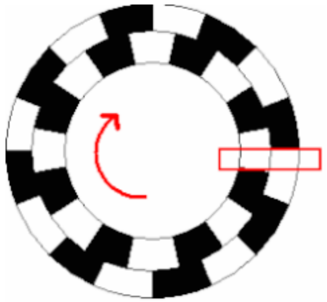
Quadrature optical wheel encoder: The observed phase relationship between channel A and B pulse trains are used to determine the direction of the rotation. A single slot in the outer track generates a reference (index) pulse per revolution.

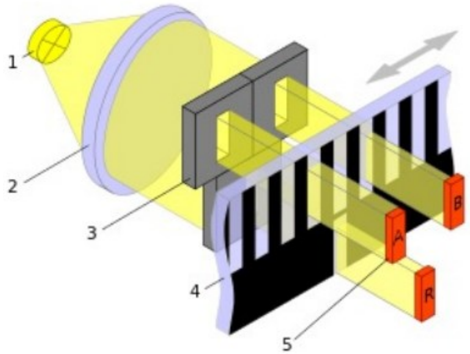
Mobile and Autonomous Robots
Optical Encoders

- The encoder disk or strip is divided into small sections and each section is either opaque or transparent.
- A light source, such as an LED on one side, projects a beam of light onto the other side of the encoder disk or strip, where it is seen by a light-sensitive sensor, such as a phototransistor.
- If the disk’s angular position (or in the case of a strip, the linear position) is such that the light is revealed, the sensor on the opposite side is turned on and has a high signal.
- If the angular position of the disk is such that the light is occluded, the pick-up sensor is off and its output is low (therefore, a digital output).
- As the disk rotates, it continuously sends signals. If the signals are counted, the approximate total displacement of the disk can be measured at any point.



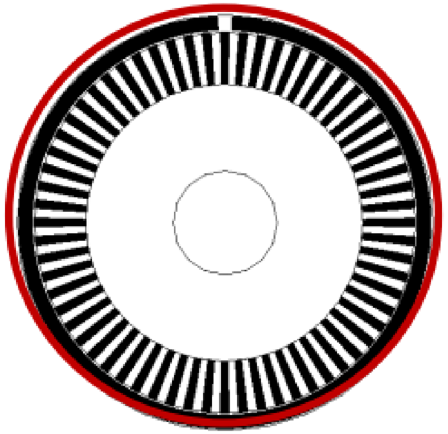






Robot moving forward and backward with Optical Position Encoder.

Given, Wheel diameter: 5.1cm, Number of slots on encoder disc: 30, find resolution.



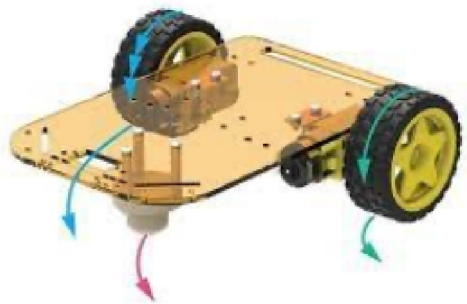
Solution:

- **Given Wheel diameter: 5.1cm**
- **Wheel circumference: 5.1cm * 3.14 = 16.014cm = 160.14mm**
- **Number slots on the encoder disc: 30**
- **Position encoder resolution: 160.14 mm / 30 = 5.34mm / pulse**

Optical Encoders: Example 2

- **Robot is turning with one wheel rotating clockwise while other wheel is Stationary. What is Position encoder resolution in degrees?**

Distance between Wheels = 15 cm.



Turning/Rotation occurs when the wheels move at different/opposite speed

- Distance between Wheels = 15cm
- Distance Covered by Robot in 360 deg. rotation =

  = Circumference of Circle traced = 2 x 15 x 3.14 = 94.20 cm

- Number of wheel rotations of in 360 deg. rotation of robot =

  = Circumference of Traced Circle / Circumference of Wheel = 942/ 160.14 = 5.882

- Total pulses in 360^0 Rotation of Robot =

  = Number of slots on the encoder disc * Number of wheel rotations of in 360 deg rotation of robot = 30 x 5.882 = 176.46 (approximately 176)

- Position Encoder Resolution in Degrees = 360 / 176= 2.040 degrees per count

Heading Sensors

- Heading sensors are used to determine the robot's orientation and inclination.
- They can be *proprioceptive* (gyroscope, inclinometer, accelerometer) or *exteroceptive* (compass).
- They allow us to integrate the movement to a position estimate. Along with velocity, the information collected can be used for position estimation.
- This procedure, which has its roots in vessel and ship navigation, is called ***dead reckoning***.
- In navigation, dead reckoning is the process of calculating the current position of a moving object by using a previously determined position, or fix, and incorporating estimates of speed, heading (or direction or course), and elapsed time.

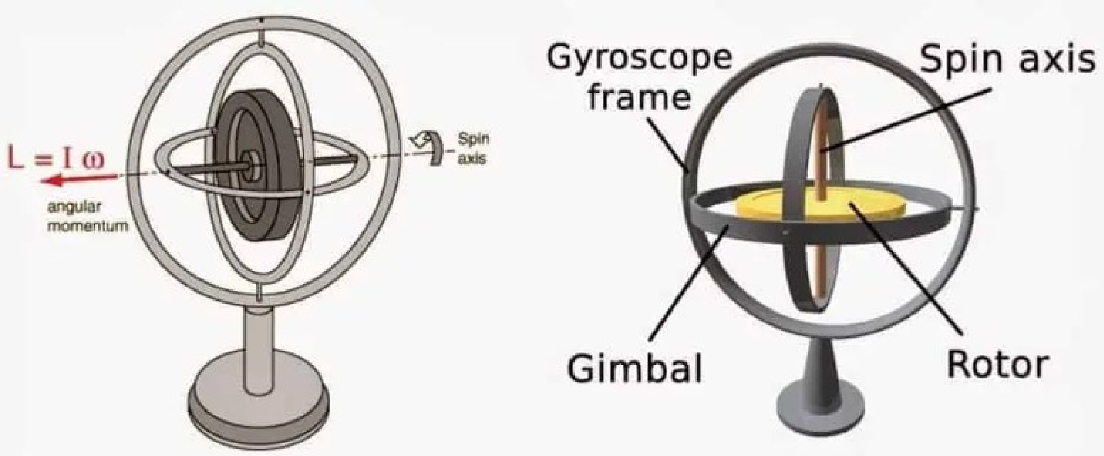


Fig. Gyroscope

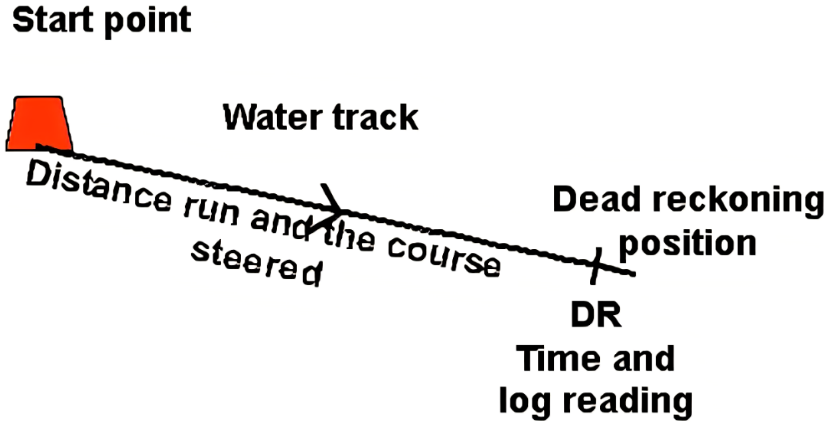


Fig. Dead Reckoning

Gyroscope

- Gyroscopes are heading sensors which measure, maintain and preserve the angular velocity and orientation of an object in relation to a fixed reference frame.
- They provide an *absolute* measure for the heading of a mobile system.
- They can measure the tilt and lateral orientation of the object.
- Gyroscope sensors are also called as Angular Rate Sensor or Angular Velocity Sensors.
- These sensors are installed in the applications where the orientation of the object is difficult to sense by humans.
- They can be classified in two categories, ***mechanical* gyroscopes** and ***optical* gyroscopes**.

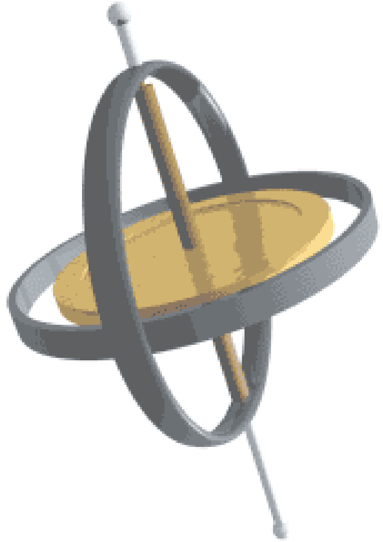
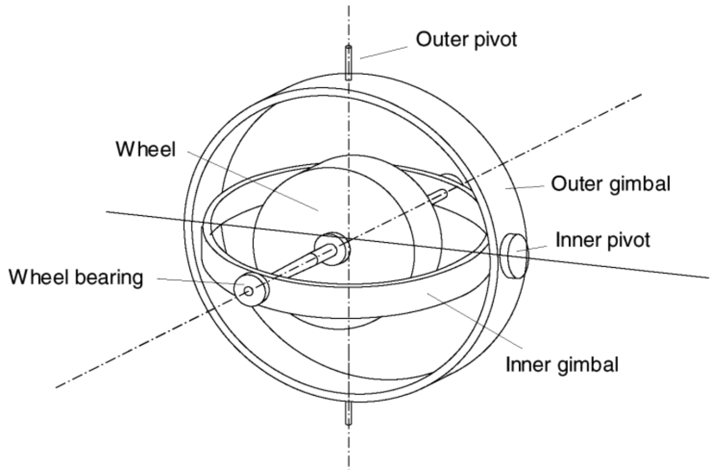


Fig. Gyroscope

Mechanical Gyroscope

- The concept of a mechanical gyroscope relies on the inertial properties of a fast-spinning rotor.
- The property of interest is known as the gyroscopic pre-cession.
- If you try to rotate a fast-spinning wheel around its vertical axis, you will feel a harsh reaction in the horizontal axis.
- This is due to the angular momentum associated with a spinning wheel and will keep the axis of the gyroscope inertially stable.



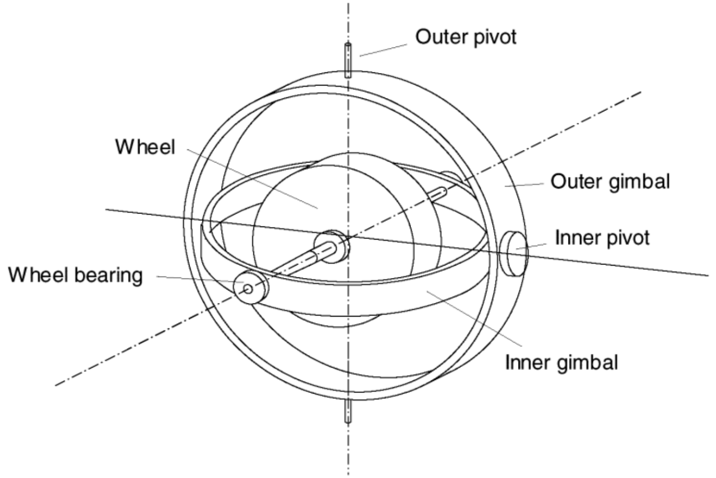
Two-axis mechanical gyroscope.

Mechanical Gyroscope

- The reactive torque τ and thus the tracking stability with the inertial frame are proportional to the spinning speed , the precession speed Ω, and the wheel's inertia *I*.

$$\tau = I\omega\Omega$$

- The angular momentum associated with a spinning wheel keeps the axis of the gyroscope inertially stable.
- For navigation, the spinning axis has to be initially selected.
- If the spinning axis is aligned with the north-south meridian, the earth's rotation has no effect on the gyro's horizontal axis.
- If it points east-west, the horizontal axis reads the earth rotation.



Two-axis mechanical gyroscope.

Optical Gyroscope

- Optical gyroscopes are angular speed sensors that use two monochromatic light beams, or lasers, emitted from the same source, instead of moving, mechanical parts.
- They work on the principle that the speed of light remains unchanged and, therefore, geometric change can cause light to take a varying amount of time to reach its destination.
- One laser beam is sent traveling clockwise through a fibre while the other travels counterclockwise.
- Because the laser traveling in the direction of rotation has a slightly shorter path, it will have a higher frequency.
- The difference in frequency Δf of the two beams is a proportional to the angular velocity Ω of the cylinder.
- New solid-state optical gyroscopes based on the same principle are build using microfabrication technology, thereby providing heading information with resolution and bandwidth far beyond the needs of mobile robotic applications.
- Bandwidth, for instance, can easily exceed 100kHz while resolution can be smaller than 0.0001degrees/hr.

Gyroscope

Accelerometer

- An accelerometers is a device used to measure all **external forces** acting upon it, including gravity.
- Accelerometers belong to the **proprioceptive** sensors class.
- Assume that an external force is applied on the sensor casing (e.g., gravity) and that we have an ideal spring with a force proportional to its displacement. Then, we can write

$$F_{applied} = F_{inertial} + F_{damping} + F_{spring} = m\ddot{x} + c\dot{x} + kx \,,$$

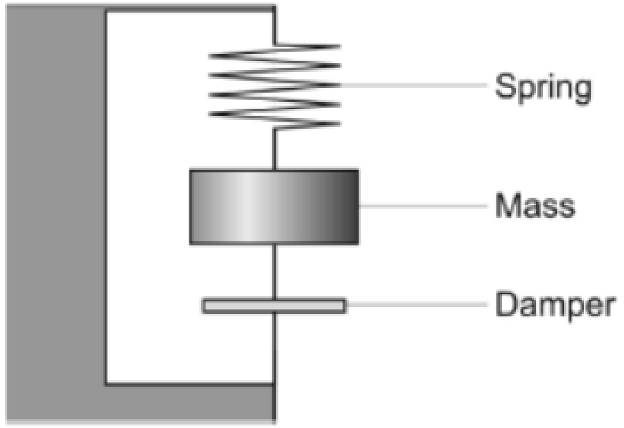
- Where m is the proof mass, c is the damping coefficient, k is the spring constant, and x is the equilibrium case relative position.

Accelerometer

- On the Earth's surface, the accelerometer always indicates 1g along the vertical axis
- To obtain the inertial acceleration (due to motion alone), the gravity must be subtracted. Conversely, the device's output will be zero during free fall
- Bandwidth up to 50 KHz
- An accelerometer measures acceleration only along a single axis. By mounting three accelerometers orthogonally to one another, a three-axis accelerometer can be obtained

No Force Applied

Inside The Accelerometer



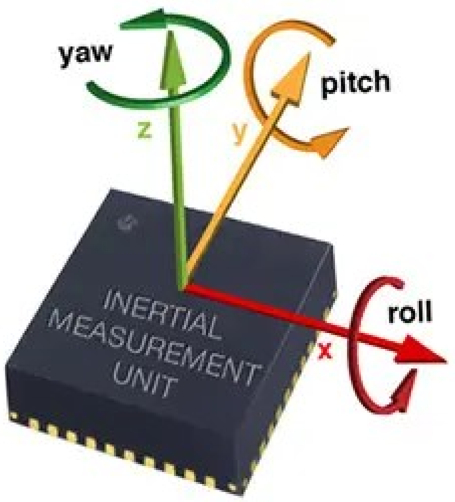
Importance of Accelerometer in Robotics

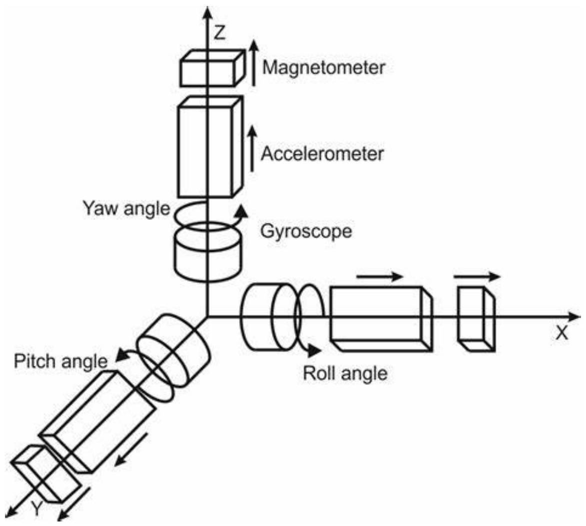
- **Motion Detection**: Accelerometers allow robots to detect and measure their own *movement and changes in velocity*. This information is crucial for **navigation, tracking, and controlling** the robot's motion.
- **Orientation Estimation**: By integrating acceleration data over time, accelerometers can also provide information about the robot's *orientation or tilt relative* to the Earth's gravity. This helps robots maintain **stability and adjust their behavior accordingly**.
- **Impact Detection**: Accelerometers can detect *sudden changes in acceleration*, which may indicate *collisions or impacts*. This information can be used to trigger **safety measures or adjust the robot's behavior** to avoid damage.
- **Gesture Recognition**: In humanoid or wearable robotics applications, accelerometers can be used to *recognize gestures or movements* performed by the user, enabling intuitive **interaction** with the robot.

Applications:

- **Inertial Measurement Units (IMUs):** Accelerometers are often integrated into IMUs along with gyroscopes and sometimes magnetometers to provide comprehensive motion sensing capabilities.
- **Robot Navigation:** Accelerometers are used in conjunction with other sensors like gyroscopes and wheel encoders to enable dead reckoning navigation, where the robot estimates its position based on its previous movement.
- **Stability Control:** In drones and other aerial vehicles, accelerometers help maintain stability by providing feedback to adjust motor speeds and control surfaces.
- **Fall Detection:** Accelerometers are used in assistive robotics and healthcare applications to detect falls or abnormal movements, triggering alerts or assistance when needed.

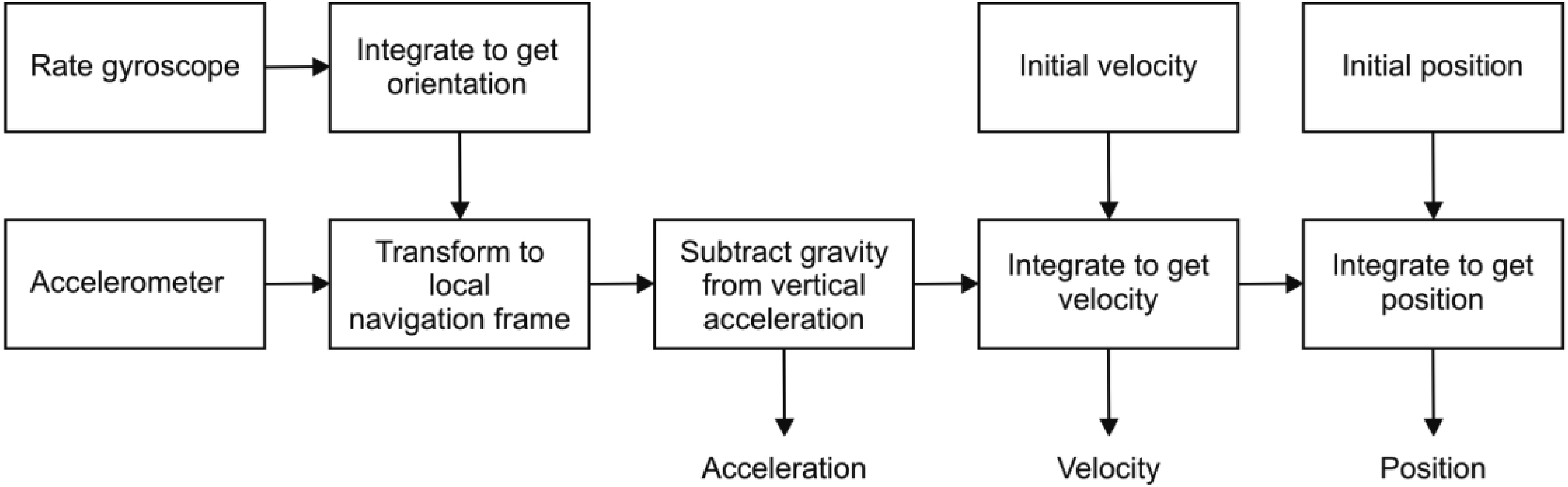
Inertial Measurement Unit (IMU)





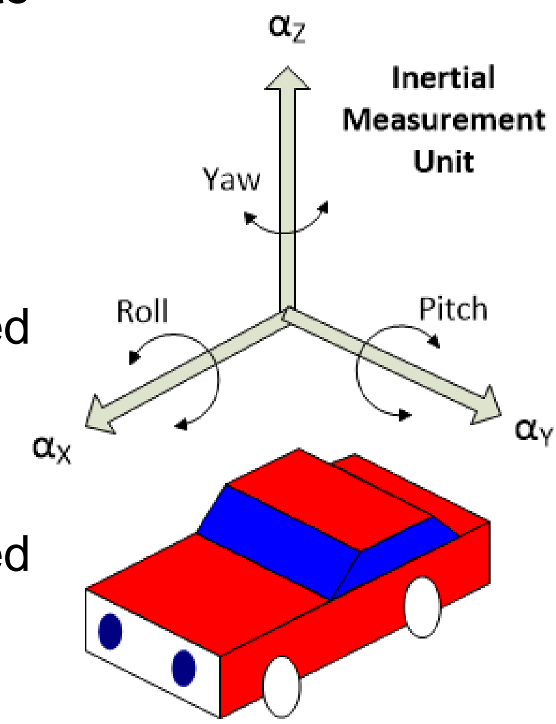
- An **inertial measurement unit (IMU)** is a device that uses *gyroscopes* and *accelerometers* to estimate the relative position, orientation, velocity, and acceleration of a moving vehicle.
- An IMU is also known as an **Inertial Navigation System (INS)**, and it has become a common navigational component of aircraft and ships. An IMU estimates the six-degree-of-freedom (6DOF) pose of the vehicle: position (x, y, z) and orientation (roll, pitch, yaw).
- Heading sensors like ***compasses and gyroscopes***, which conversely **only estimate orientation**, are often improperly called IMUs.

IMU block diagram



Inertial Measurement Unit (IMU)

- Commercial IMUs also usually estimate velocity and acceleration. To estimate the velocity, the initial speed of the vehicle needs to be known.
- Gyroscope data is used to estimate vehicle orientation.
- Accelerometers estimate instantaneous acceleration, which is then transformed to the local navigation frame based on the current orientation relative to gravity.
- The resulting acceleration is integrated to obtain velocity and then integrated again to obtain position, starting from rest (zero initial velocity).

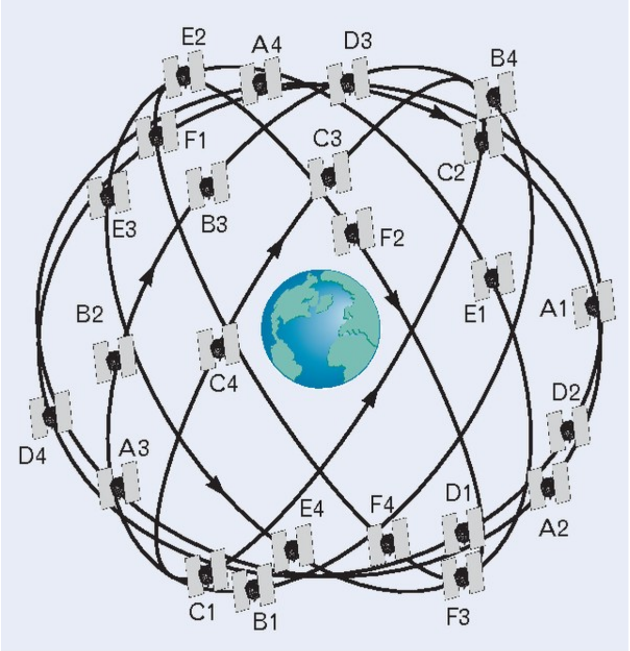


Inertial Measurement Unit (IMU)

- IMUs are highly sensitive to measurement errors from both gyroscopes and accelerometers.
- Gyroscope drift undermines accurate estimation of vehicle orientation relative to gravity, leading to incorrect cancellation of the gravity vector.
- Accelerometer data, integrated twice to calculate position, amplifies any residual gravity vector error, resulting in quadratic error in position estimation.
- Drift is inherent in IMUs and accumulates over time due to integration of errors.
- To mitigate drift, external reference measurements from sources like cameras or GPS are used. Cameras can correct drift by observing known features in the environment, while GPS signals allow for periodic correction of pose estimates.

Global Positioning Systems (GPS)

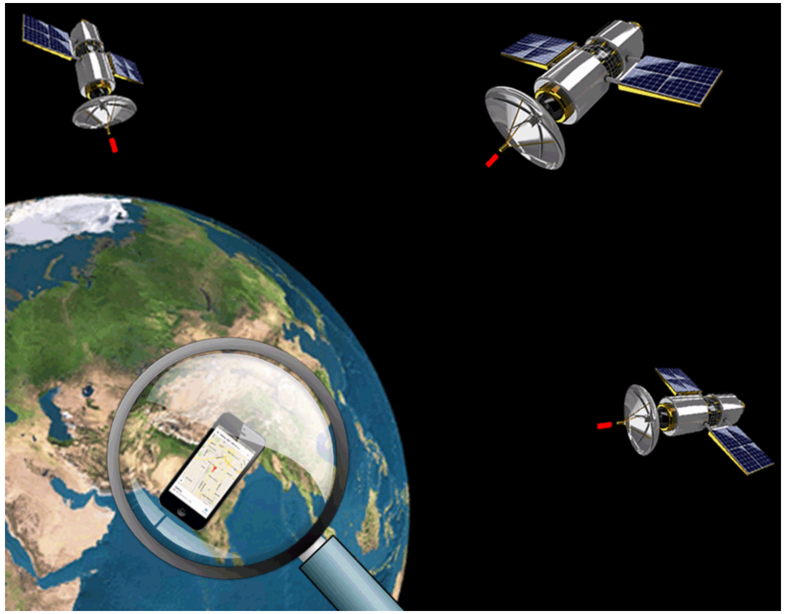
- The **global positioning system (GPS)** was initially developed for military use but is now freely available for civilian navigation.
- GPS units are generally used in mobile robots and navigation systems to determine the current position of the device relative to a global reference frame.
- With a GPS receiver, we can determine a global position and time that can be used for navigation and mapping.
- The system includes many satellites orbiting the Earth at about 20,000 km, a control and monitoring station on Earth, and the GPS receivers.
- There are always at least 24 GPS satellites. The satellites orbit every 12 hours.
- Four satellites are in each of six planes inclined 55 degrees with respect to the plane of the earth’s equator.



- The receiver uses the transmitted data from the satellites to calculate its position.
- This information can be sent directly to the control system of a mobile robot for positioning purposes and navigation.
- Therefore, GPS receivers are **completely passive but exteroceptive** sensors.
- The GPS satellites synchronize their transmissions so that their signals are sent at the same time.
- When a GPS receiver reads the transmission of two or more satellites, the arrival time differences inform the receiver as to its relative distance to each satellite.
- By combining information regarding the arrival time and instantaneous location of four satellites, the receiver can infer its own position.

Mobile and Autonomous Robots
Global Positioning Systems (GPS)

- Triangulation, based on the arrival time differences of signals from multiple satellites, enables the receiver to calculate its position.
- Ideally, triangulation requires signals from at least three satellites, but four are typically used to account for timing accuracy.
- Timing is crucial in GPS applications, with measurements conducted in nanoseconds, necessitating synchronization of satellite transmissions.
- The satellites are well synchronized, updated by ground stations regularly and each satellite carries on-board atomic clocks for timing.
- The GPS receiver clock is also important so that the travel time of each satellite's trans-mission can be accurately measured.



Limitations of GPS

- The fact that the GPS receiver must read the transmission of four satellites simultaneously is a significant limitation. GPS satellite transmissions are extremely low-power, and requires direct line-of-sight communication with the satellite.
- Confined spaces such as city blocks with tall buildings or in dense forests, one is unlikely to receive four satellites reliably.
- Most indoor spaces will also fail to provide sufficient visibility of the sky for a GPS receiver to function.
- Due to these limitations, GPS is commonly used in mobile robotics projects, particularly in traversing wide-open spaces and by autonomous flying machines.
- However, GPS is less effective in environments with obstructed visibility, such as urban areas or dense foliage.
- The reliance on direct line-of-sight communication restricts the applicability of GPS in certain scenarios.
- Despite its limitations, GPS remains a valuable sensor for outdoor navigation, where it provides accurate positioning and timing information.

Factors affecting performance of localization sensor

- The coverage and resolution of GPS-based localization sensors vary depending on the orbital paths of GPS satellites. At the poles, where satellites are close to the horizon, altitude resolution is relatively poor compared to equatorial regions.
- Different strategies can be employed with GPS to achieve varying levels of localization resolution. The basic pseudorange method offers a resolution of around 15 meters.
- Differential GPS (DGPS) improves resolution to approximately 1 meter or less by using a second stationary receiver at a known position to correct errors.
- DGPS requires careful installation of the stationary receiver and proximity of the moving robot to the stationary unit, limiting its practicality.
- Advanced techniques involving carrier signal phase measurement can achieve resolutions as fine as 1 centimeter for point positions, and even sub-1 centimeter with multiple receivers.
- GPS updates typically have latency of 200 to 300 milliseconds, resulting in updates at a maximum rate of 5 Hz. Fast-moving robots may require motion integration to compensate for GPS latency during control.

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Mobile and Autonomous Robots (UE23CS343BB7)

PERCEPTION

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Mobile and Autonomous Robots

UE23CS343BB7

Perception in Autonomous Robots

Perception Sensors: Range sensors, camera vision and LiDAR

Non-Perception Sensors

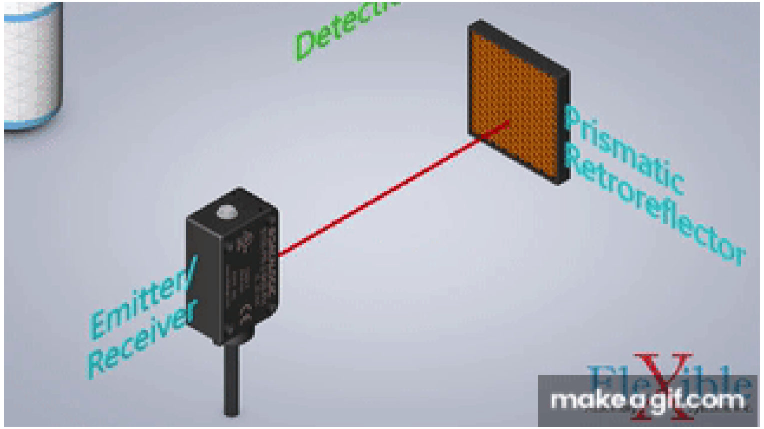
- Encoders
- Gyroscope
- Accelerometer
- IMU (Inertial Measurement Unit)
- GPS / GNSS / RTK-GPS

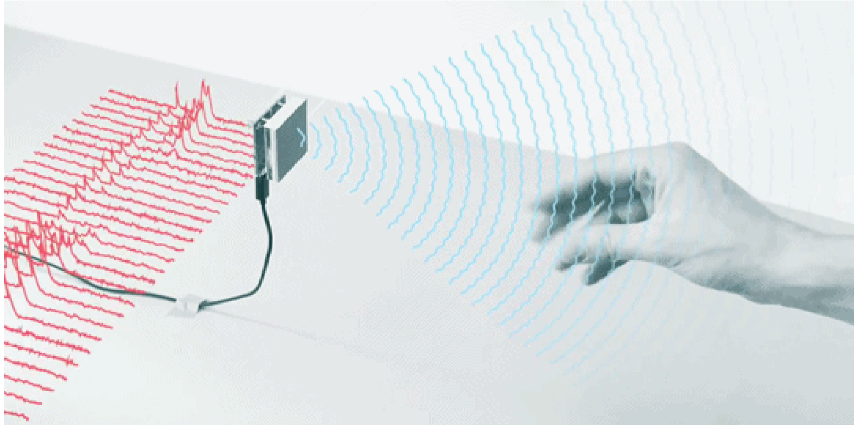
Perception Sensors

- LiDAR (2D / 3D)
- RADAR
- Ultrasonic Sensors
- Infrared / Thermal Cameras
- RGB / Stereo / Depth Cameras

Range Finders

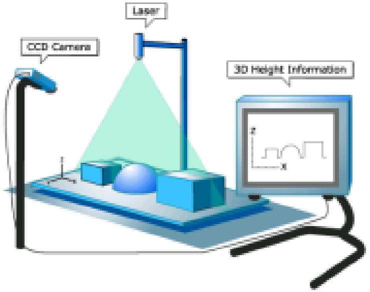
- Used for direct measurements of **distance** from the robot to objects in its vicinity.
- For obstacle detection and avoidance, most mobile robots rely heavily on active ranging sensors.
- Used to find larger distances, to detect obstacles, and to map the surfaces of objects.
- Generally based on light--**visible light, infrared light**, or **laser--and ultrasonic.**
- Two common methods of measurement are **triangulation and time-of-flight** or lapsed time.
- Active ranging sensors are also commonly found as part of the localization and environmental modelling processes of mobile robots.





Active Ranging

- SONAR
- Laser range finder
- Time of flight camera
- Structured light



Structured Light

Laser range finder

- Make use of wave propagation to measure large range distance measurement
- Ultrasonic sensors as well as laser range sensors make use of propagation speed of sound or electromagnetic waves respectively

Ultrasonic

$$d = ct$$

d: distance, c: speed of signal, t: time of flight

Time of Flight

- Time-of-flight ranging makes use of the propagation speed of sound or an electromagnetic wave.

- The quality of time-of-flight range sensors depends mainly on
  - uncertainties in determining the exact time of arrival of the reflected signal;
  - inaccuracies in the time-of-flight measurement
  - the dispersal cone of the transmitted beam
  - interaction with the target (e.g., surface absorption, specular reflections);
  - variation of propagation speed;
  - the speed of the mobile robot and target (in the case of a dynamic target);

The ultrasonic sensor (time-of-flight, sound)

The basic principle of an **ultrasonic sensor** is to transmit a packet of (ultrasonic) **pressure waves** and to measure the time it takes for this wave packet to reflect and return to the receiver.

The distance *d* of the object causing the reflection can be calculated based on the propagation speed of sound *c* and the time of flight *t*.

$$d = \frac{c.t}{2}$$

The speed of sound c in air is given by

where ,

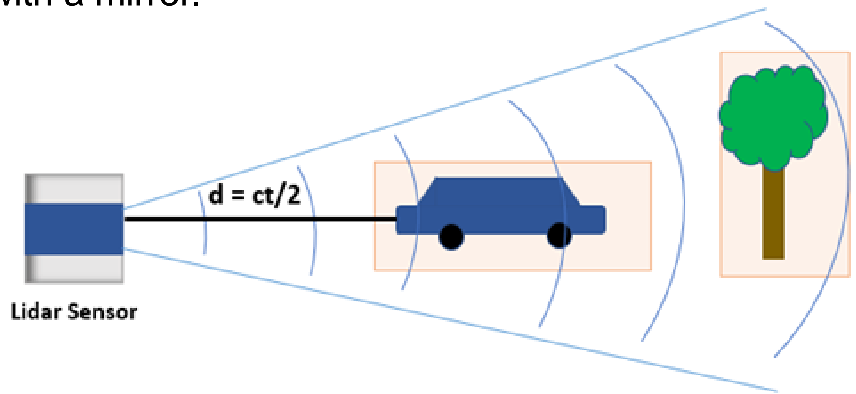
$$c = \sqrt{\gamma RT}$$

γ = ratio of specific heats, *R* = gas constant, *T* = temperature in Kelvin

In air at standard pressure and 20° C the speed of sound is approximately = 343 m/s.

Laser rangefinder (time-of-flight, electromagnetic)

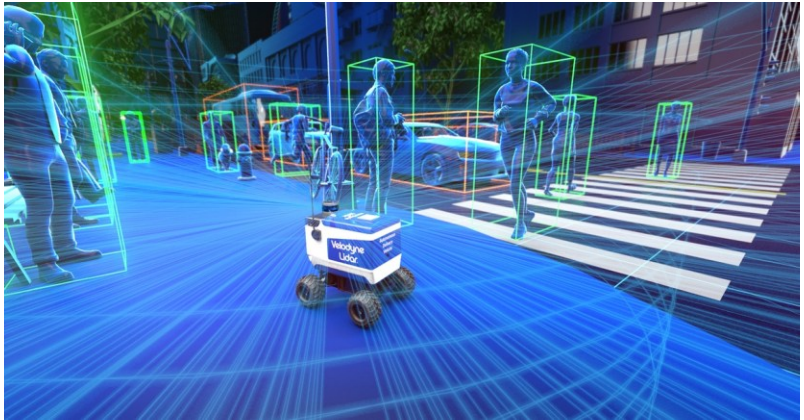
- **Laser rangefinders**, also known as optical radar or lidar, use **laser light** instead of sound for distance measurement.
- These sensors consist of a transmitter that emits a collimated beam of laser light and a receiver capable of detecting the reflected light component.
- Sensors produce range estimates based on the time it takes for the light to travel to the target and return.
- They can cover the required scene in either a plane or three dimensions using a mechanical mechanism with a mirror.

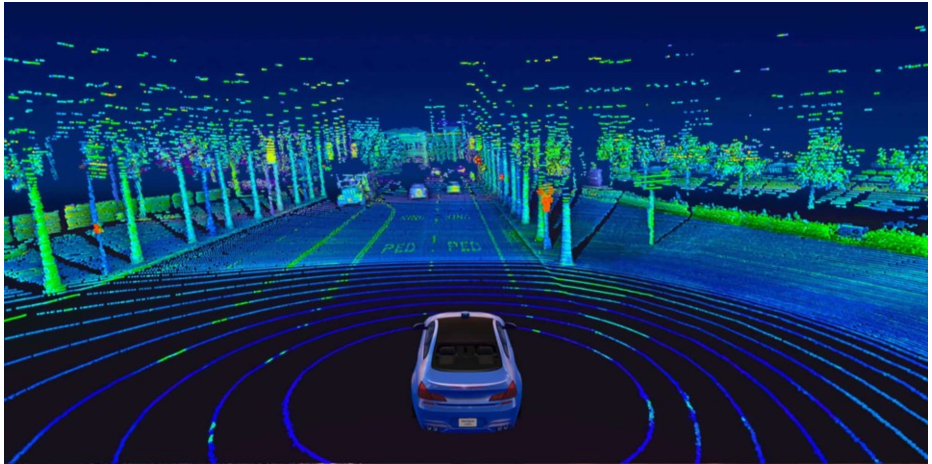


d = distance between point and the sensor
c = speed of light
t = time of flight

LiDAR: Light Detection And Ranging

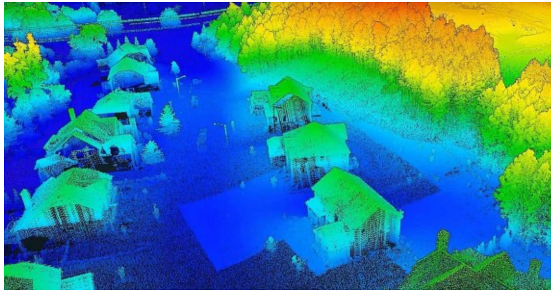
- Lidar is a method for determining ranges by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the receiver.
- It is a detection system which works on the principle of radar but uses light from a laser.
- It shoots pulses of laser light and measures the time it takes to bounce back from objects.
- Robots use lidar for tasks like navigation, obstacle avoidance, object recognition, and even self-driving.

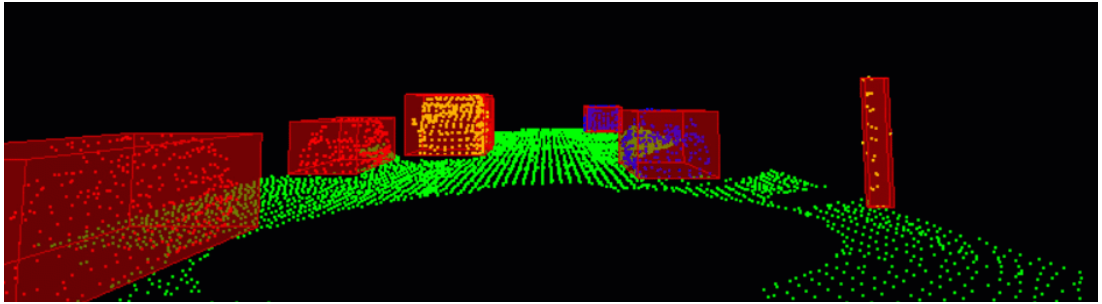




LiDAR: Light Detection And Ranging

- Lidar creates detailed 3D maps of surroundings, allowing robots to navigate autonomously, avoid obstacles, and plan efficient paths.
- By analysing the reflected light, robots can identify and classify objects in their environment, enabling tasks like picking and placing items or recognizing people.
- Lidar helps robots build a map of their environment while simultaneously tracking their own location within it, and detects obstacles in real-time, allowing robots to safely navigate dynamic environments and avoid collisions.

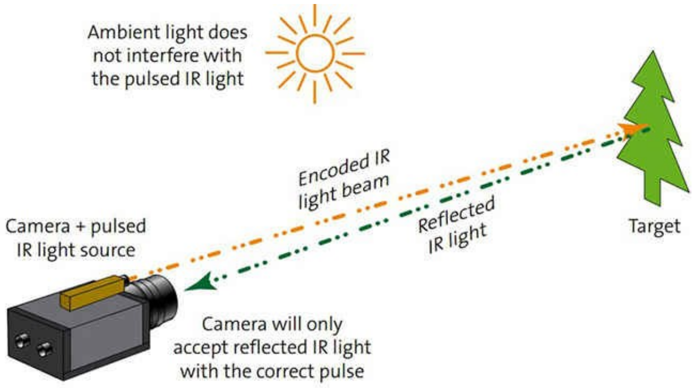




Mobile and Autonomous Robots

Time-of-flight camera

- A Time-of-Flight (TOF) camera functions similarly to a lidar but captures the entire 3D scene simultaneously without any moving parts.
- It utilizes a modulated **infrared lighting source** to determine the distance for each pixel of a Photonic Mixer Device (PMD) sensor.
- The compact design of TOF cameras, with the illumination source placed next to the lens, distinguishes them from lidars, stereo vision, or triangulation sensors.
- To eliminate background light interference, the acquisition is performed twice: once with illumination and once without illumination, allowing for up to 100 frames per second and making the camera suitable for real-time applications.



Triangulation active ranging

- Triangulation ranging sensors utilize geometric properties to determine distance readings to objects.
- Active triangulation sensors project a known light pattern onto the environment and capture the reflection with a receiver.
- By combining known geometric values with the captured reflection, the sensor can use triangulation to establish range measurements.
- **Optical triangulation** sensors measure the position of the reflection along a single axis, while **structured light** sensors measure the position along two orthogonal axes.

Triangulation active ranging

The 2 sensor types are :

- Optical triangulation (1D sensor).
- Structured light (2D sensor)

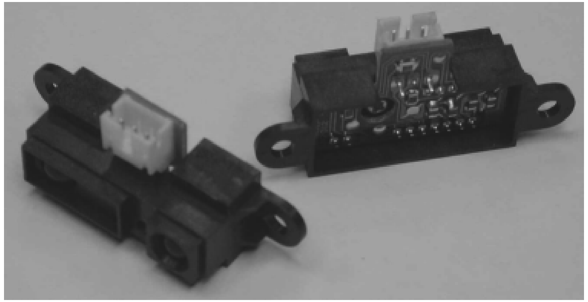


Figure 4.22
A commercially available, low-cost optical triangulation sensor: the Sharp GP series infrared rangefinders provide either analog or digital distance measures and cost only about $15.

Optical triangulation (1D sensor)

A collimated beam (e.g., focused infrared LED, laser beam) is transmitted toward the target. The reflected light is collected by a lens and projected onto a position-sensitive device (PSD) or linear camera. Given the geometry, the distance *D* is given by

$$D = f\frac{L}{x}.$$

Triangulation active ranging

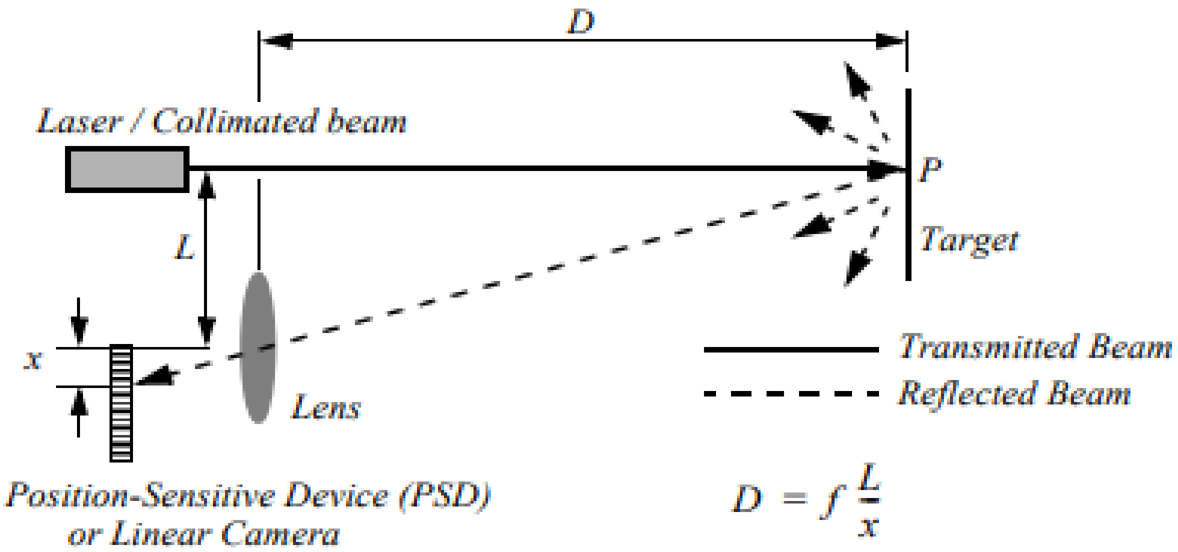


Figure 4.21
Principle of 1D laser triangulation.

Optical triangulation (1D sensor)

Triangulation active ranging

Structured light (2D sensor)

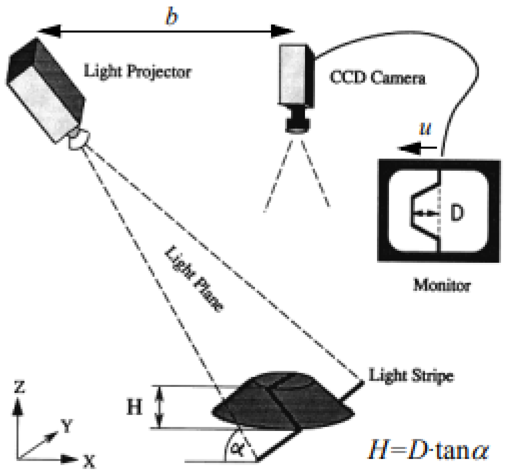
- Structured light sensors, also known as 2D sensors, replace the linear camera or PSD of an optical triangulation sensor with a 2D receiver like a CCD or CMOS camera.
- These sensors enable the recovery of distance to a large set of points instead of just one point.
- The emitter of structured light sensors projects a known pattern, such as light textures or laser stripes, onto the environment.
- Light textures or laser stripes can be generated by various methods, including projecting light textures or emitting collimated light using a rotating mirror, or projecting a laser stripe using a prism.

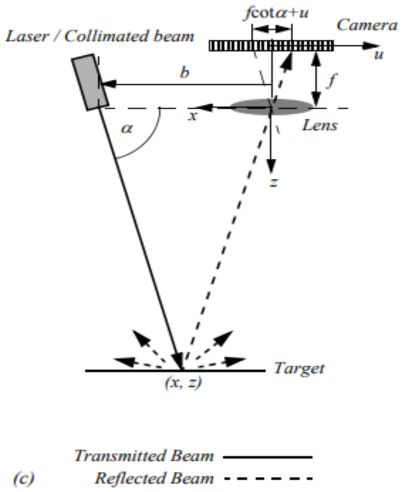
Triangulation active ranging

- The projected light pattern has a known structure, allowing the image taken by the CCD or CMOS receiver to be filtered to identify the pattern's reflection.
- Structured light sensors simplify the problem of recovering depth compared to passive image analysis, as they project a known pattern onto the environment, avoiding the need for correlation with existing features.
- Structured light sensors are active devices, functioning in dark environments and featureless surroundings where passive image analysis or stereo vision would fail.
- They are effective in environments with uniform color and no distinct edges, providing accurate depth measurements.

Triangulation active ranging

- The use of structured light sensors offers advantages over passive image analysis and stereo vision, particularly in featureless or low-texture environments.
- Structured light sensors are widely used in various applications where precise depth measurement and imaging are required, such as 3D scanning, robotics, and augmented reality.

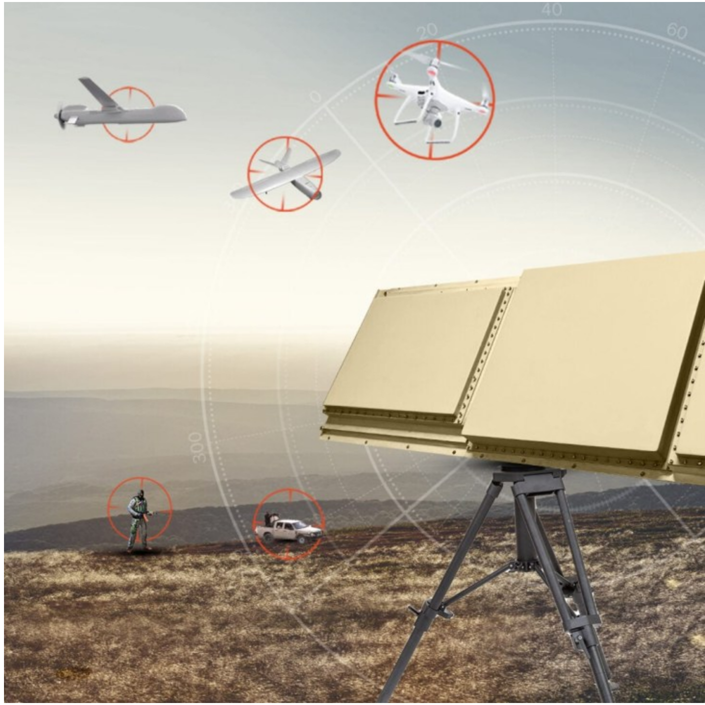




RADAR

RADAR stands for **R**adio **D**etection **A**nd **R**anging

- It is a sensing system used to **detect, locate, and track objects** by using **electromagnetic (radio) waves**.
- What Radar Does
    - Detects the presence of objects
    - Measures the distance (range)
    - Determines speed (via Doppler shift)
    - Estimates direction and angle



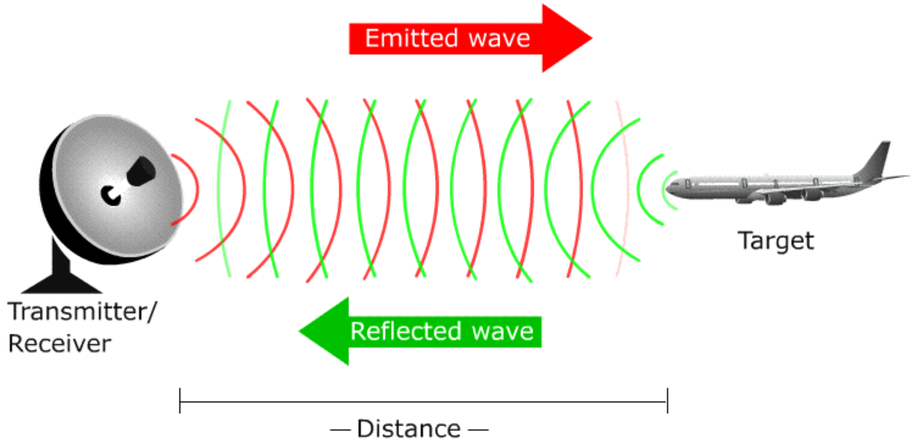
RADAR

How Radar Works

1. A **transmitter** emits radio waves.
2. The waves **hit an object** (target).
3. The object **reflects (echoes)** the waves back.
4. A **receiver** captures the reflected signal.
5. The **time delay** between transmission and reception is used to calculate distance.

$$\text{Distance} = \frac{c \times \textit{time delay}}{2}$$

(where c is the speed of light)

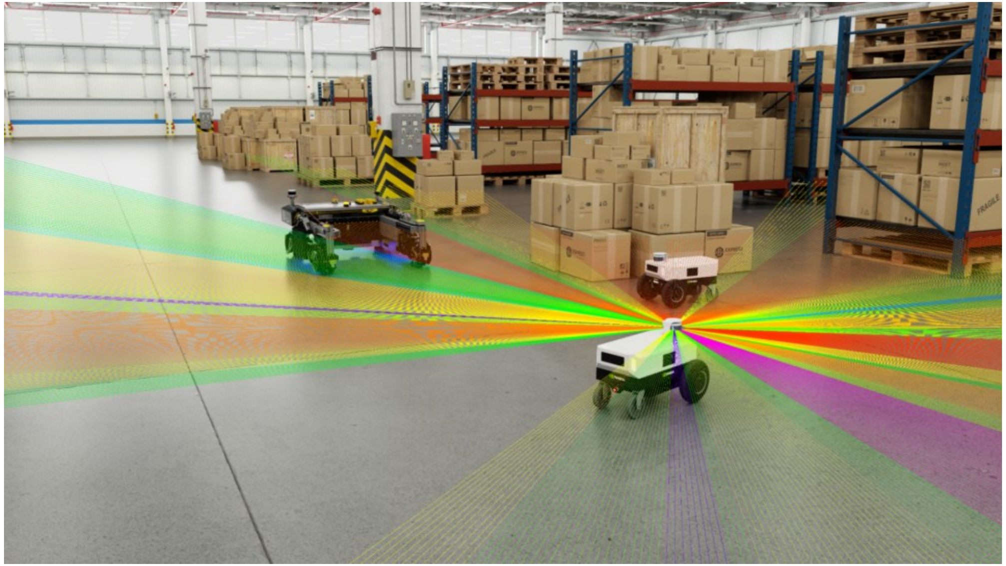


Mobile and Autonomous Robots

RADAR

Key Characteristics

- Detects objects using radio waves
- Works in darkness and fog
- Measures distance and speed
- Robust in harsh environments
- Complements cameras and LiDAR



RADAR

Applications of Radar in Robotics

- Autonomous Navigation
  - Radar helps robots detect obstacles and navigate safely in complex environments, especially where visibility is limited or lighting conditions are conditions are poor.
- Object Detection and Tracking
  - Robotic systems use radar to track moving objects, estimate their speed, and speed, and predict motion, which is essential for collision avoidance.
- Human-Robot Interaction
  - Radar sensors can detect human presence and movement, enabling safer and enabling safer and more responsive interaction between robots and people and people.

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Mobile and Autonomous Robots

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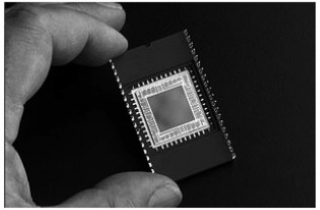
Robot Vision

Fundamentals of Computer Vision

Computer Vision

- Computer vision is a field that includes methods for acquiring, processing, analyzing, and understanding images
- Known as Image analysis, Scene Analysis, Image Understanding
- Duplicate the abilities of human vision by electronically perceiving and understanding an image
- Theory for building artificial systems that obtain information from images.
- Image data can take many forms, such as a video sequence, depth images, views from multiple cameras, medical scanner, satellite sensors etc.

- Vision is our most powerful sense.
- It provides us with an enormous amount of information about the environment and enables rich, intelligent interaction in dynamic environments.
- The first step in this process is the creation of sensing devices that capture the same raw information light that the human vision system uses



2048 x 2048 CCD array

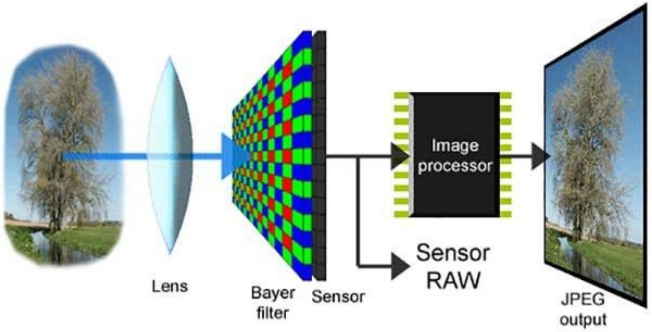
Orangemicro iBOT Firewire

Sony DFW-X700

Canon IXUS 300

The digital camera

- Light from one or more sources reflects off surfaces and passes through the camera's lenses to reach the imaging sensor.
- Photons arriving at the sensor are converted into digital (R, G, B) values, forming the basis of a digital image.
- The imaging sensor typically consists of an active sensing area where light is integrated over the exposure duration, often expressed as shutter speed.
- After integration, the sensed light is passed to a set of sense amplifiers for further processing.

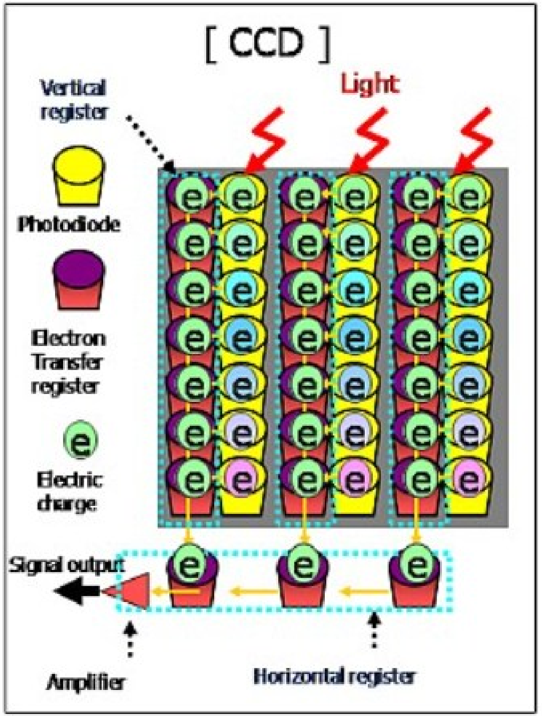


The digital camera

- Two main types of sensors used in digital cameras are **CCD (charge-coupled device)** and **CMOS (complementary metal-oxide-semiconductor)**.
- **CCD** sensors offer advantages such as high image quality and low noise but may consume more power and be slower in operation.
- **CMOS** sensors have advantages in terms of lower power consumption, faster operation, and potential for on-chip processing, but may exhibit higher noise levels.
- Both sensor types have their own strengths and drawbacks, influencing their suitability for different applications in digital imaging.

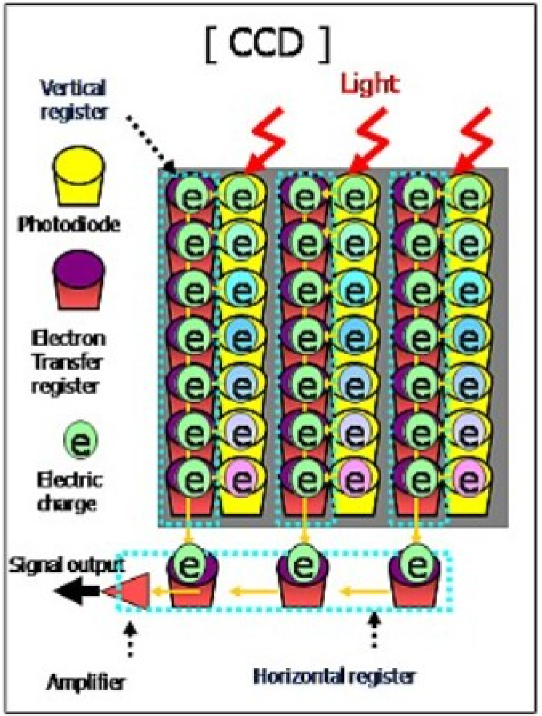
CCD cameras

- CCD chips consist of an array of light-sensitive pixels, each acting as a discharging capacitor when exposed to light.
- Photons striking each pixel liberate electrons, accumulating charge proportional to the incident light intensity during the integration period.
- Reading the relative charges of pixels involves transporting them across the chip to one corner for processing, requiring specialized control circuitry.
- Photodiodes in CCD chips are sensitive to light within a wavelength range of 400 to 1000 nm, with decreased sensitivity to ultraviolet and increased sensitivity to infrared.



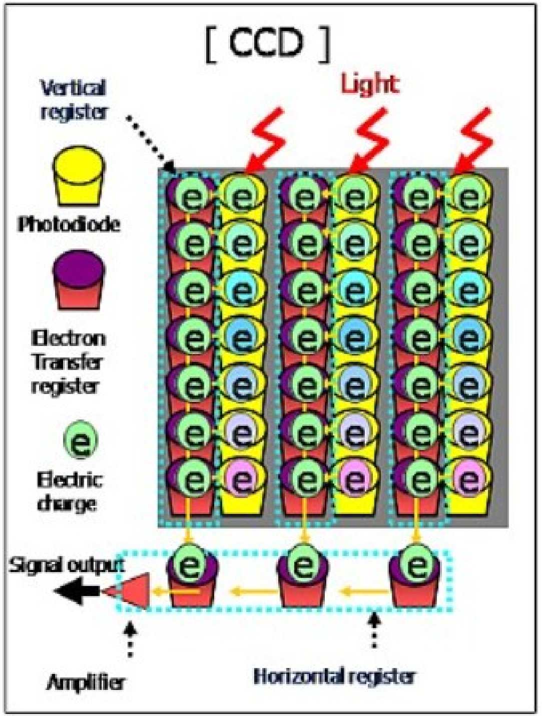
Fundamentals of Computer Vision

- CCD camera parameters include iris position, shutter speed, and camera gain, affecting light measurement and signal amplification.
- Disadvantages of CCD cameras include inconsistency in brightness and color rendition, particularly challenging in varying environments.
- CCD chips exhibit limitations in extreme illumination conditions, with low illumination leading to noise and high illumination causing pixel saturation and blooming effects.
- The dynamic range of a CCD camera is limited by the well capacity of individual pixels, typically ranging between 20,000 and 350,000 electrons, influencing the signal-to-noise ratio and overall image quality.



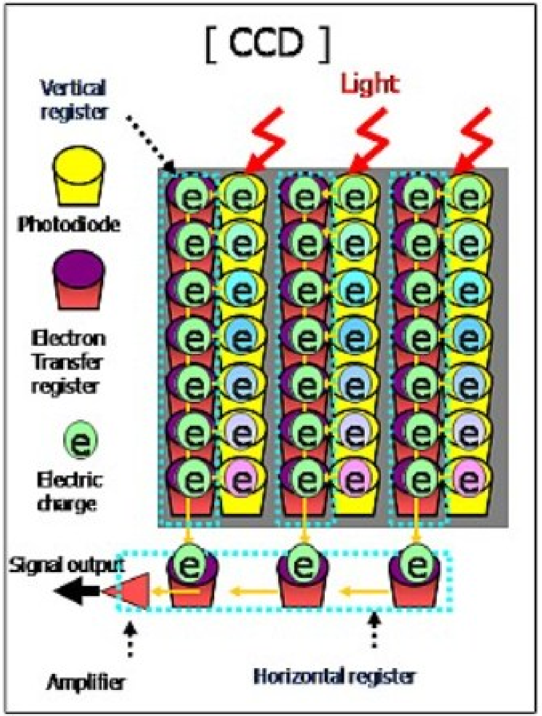
Fundamentals of Computer Vision

- The integration period of CCD chips begins after capacitors in pixels are fully charged, with photons accumulating charge proportional to light intensity.
- During the integration period, each pixel accumulates a varying level of charge based on the total number of photons it receives.
- Reading pixel charges involves shifting them to one corner of the chip for processing, requiring precise control to maintain charge stability.
- CCD cameras offer various parameters for adjusting exposure, including iris position, shutter speed, and camera gain, providing flexibility in capturing images.

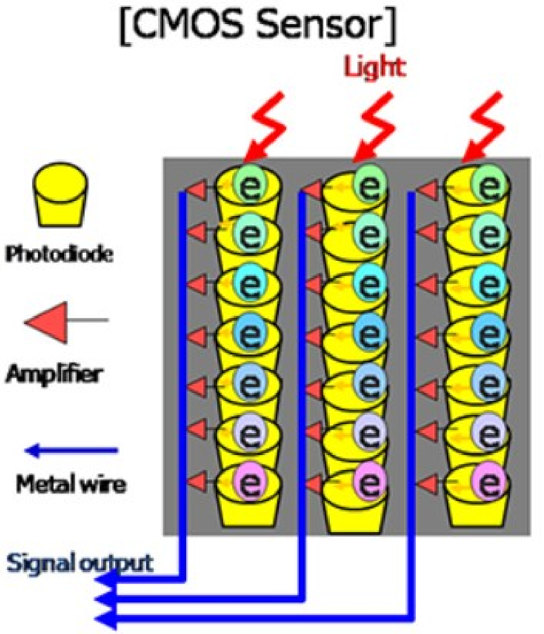


Fundamentals of Computer Vision

- Inconsistency in brightness and color rendition can arise from changes in camera parameters, challenging color constancy and luminosity constancy.
- Extreme illumination conditions can lead to pixel saturation and blooming effects, affecting image quality and dynamic range.
- The dynamic range of a CCD camera, determined by the well capacity of pixels, influences its ability to capture details in both bright and dark areas of an image.
- Noise levels during pixel reading can affect the signal-to-noise ratio, impacting image clarity and fidelity.
- CCD cameras are widely used in applications requiring high-quality imaging, such as astronomy, microscopy, and professional photography.
- Advances in sensor technology have led to improvements in CCD cameras, enhancing their performance and expanding their applications in various fields.



Fundamentals of Computer Vision

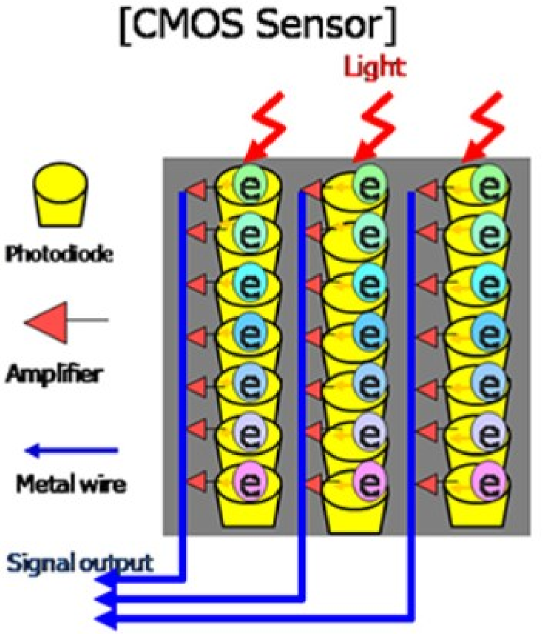


CMOS cameras

- CMOS chips feature an array of pixels with specific transistors located alongside each pixel, allowing for pixel-specific signal measurement and amplification.
- Unlike CCD chips, CMOS chips measure and amplify pixel signals in parallel for every pixel in the array during the data collection step.
- CMOS technology offers several advantages over CCD, including simplified manufacturing processes and reduced power consumption, making it more suitable for low-power applications like mobile robotics.
- The absence of specialized clock drivers and circuitry in CMOS eliminates the need for complex transfer mechanisms, further simplifying the chip design and manufacturing process.

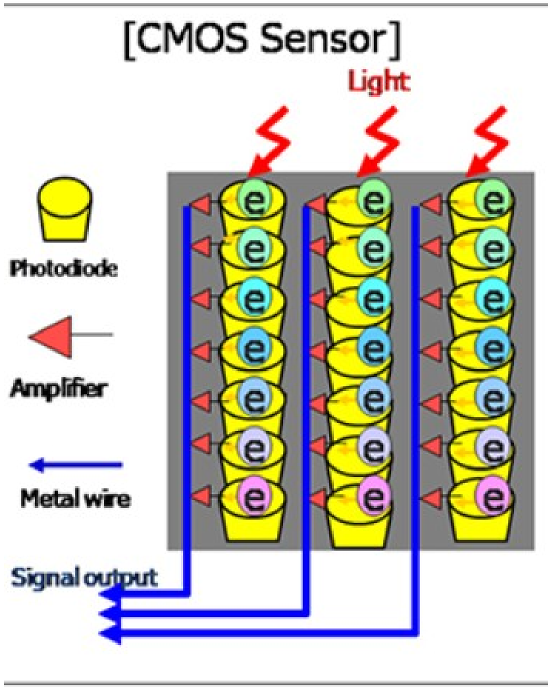
Fundamentals of Computer Vision

- CMOS chips can be produced on the same manufacturing lines as microchips, enhancing cost-effectiveness and scalability.
- While CCD sensors traditionally outperformed CMOS in quality-sensitive applications, CMOS has become predominant in most digital cameras today.
- Vision sensors based on CCD or CMOS chips exhibit limitations in adaptation, cross-sensitivity, and dynamic range compared to the human eye, making them fragile and requiring ongoing improvements in chip performance for greater robustness in mobile robot applications.



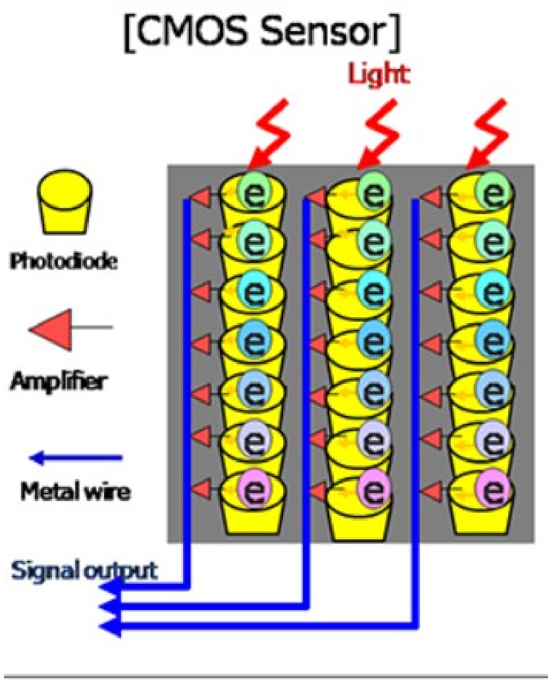
Fundamentals of Computer Vision

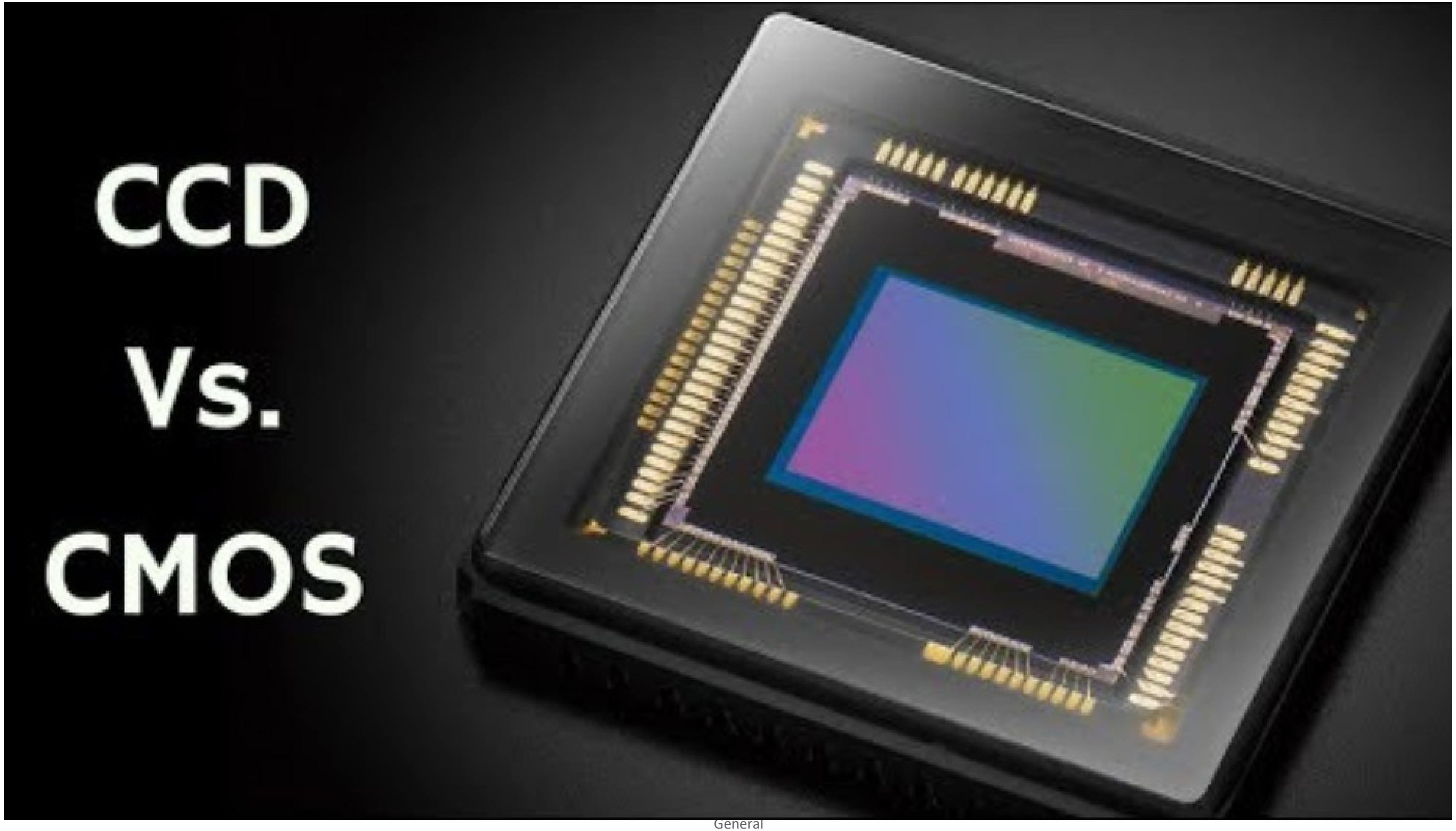
- **CMOS has several advantages over CCD technologies:**
  - There is no need for the specialized **clock drivers** and **circuitry required** in the CCD to transfer each pixel's charge down all the array columns and across all of its rows.
  - Semiconductor manufacturing processes are not required to create CMOS chips. Therefore, the same production lines that create microchips can create **inexpensive CMOS chips**.
  - The CMOS chip is so much simpler that it consumes **significantly less power**. It operates with a power consumption that is 1/100th the power consumption of a CCD chip. In a **mobile robot**, power is a scarce resource and therefore is an important advantage.



Fundamentals of Computer Vision

- **CMOS also has a few disadvantages:**
  - The circuitry next to each pixel consumes valuable real estate on the face of the light-detecting array.
  - Many photons hit the transistors rather than the photodiode, making the CMOS chip significantly less sensitive than an equivalent CCD chip.
  - The CMOS technology is younger and, as a result, the best resolution that one can purchase in CMOS format continues to be far inferior to the best CCD chips available.




CCD
Vs.
CMOS

Color camera

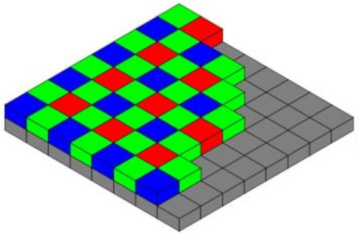
- The basic light-measuring process described before is colorless: it is just measuring the total number of photons that strike each pixel in the integration period. There are two common approaches for creating color images, which use a **single chip** or **three separate chips**.
- The **single chip** technology uses the so-called **Bayer filter**. The pixels on the chip are grouped into sets of four, then red, green, and blue color filters are applied so that each individual pixel receives only light of one color.
- Normally, two pixels of each 2 X 2 block measure green while the remaining two pixels measure red and blue light intensity.
- The reason there are twice as many green filters as red and blue is that the luminance signal is mostly determined by green values, and the visual system is much more sensitive to high frequency detail in luminance than in chrominance.

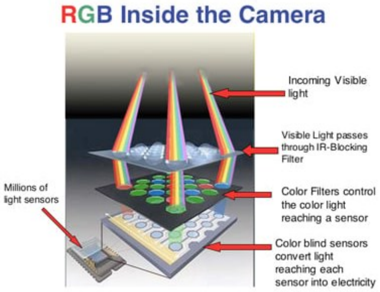
Fundamentals of Computer Vision

- The process of interpolating the missing color values so that we have valid RGB values as all the pixels is known as Demosaicing. One-chip technology has a geometric resolution disadvantage.
- The number of pixels in the system has been effectively cut by a factor of four, and therefore the image resolution output by the camera will be sacrificed.
- Three-chip color cameras split incoming light into three complete copies, with each copy directed to a separate chip equipped with red, green, or blue filters.
- Each chip measures light intensity for one color in parallel, preserving resolution while requiring the combination of outputs from all three chips to generate a joint color image.
- While three-chip color cameras maintain resolution,

  they are significantly more expensive and less commonly

  used in mobile robotics compared to single-chip cameras.





General

Fundamentals of Computer Vision

- Both three-chip and single-chip color cameras face challenges with photodiode sensitivity, particularly towards the near-infrared end of the spectrum, resulting in poorer detection of blue light.
- Compensating for the photodiode sensitivity issue requires increasing gain on the blue channel, leading to greater absolute noise on blue compared to red and green channels.
- White balance controls in color cameras allow users to adjust the relative gains for red, green, and blue to maintain consistent color definitions in varying lighting conditions, addressing issues such as color inconsistency caused by different sources of illumination.

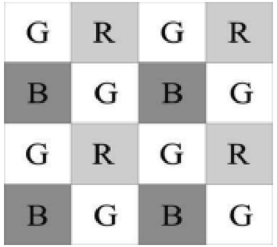


Figure 4.27 Bayer color filter array.

Image Formation Optics

According to the lens law, the relationship between the distance to an object and the distance behind the lens at which a focused image is formed can be expressed as

$$\frac{1}{f} = \frac{1}{z} + \frac{1}{e}$$

where is f the focal length. As you can perceive, this formula can also be used to estimate the distance to an object by knowing the focal length and the current distance of the image plane to the lens. This technique is called *depth from focus.*

Fundamentals of Computer Vision

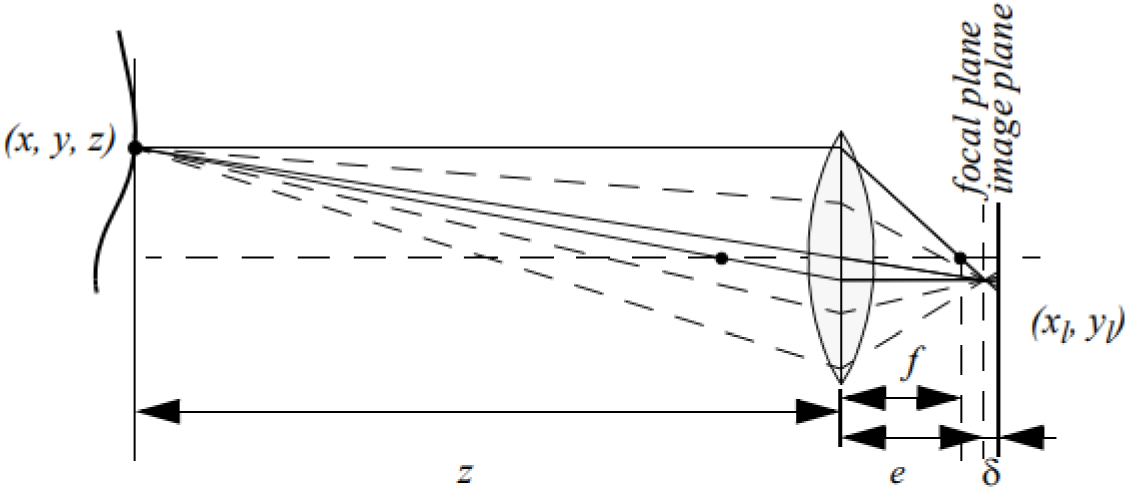


Figure 4.28
Depiction of the camera optics and its impact on the image. In order to get a sharp image, the image plane must coincide with the focal plane. Otherwise the image of the point (x,y,z) will be blurred in the image, as can be seen in the drawing above.

- If the image plane is located at distance from the lens, then for the specific object voxel depicted, all light will be focused at a single point on the image plane and the object voxel will be focused.
- However, when the image plane is not at e (as is depicted in figure 4.28), then the light from the object voxel will be cast on the image plane as a blur circle (or circle of confusion).
- To a first approximation, the light is homogeneously distributed throughout this blur circle, and the radius R of the circle can be characterized according to the equation

$$R = \frac{L\delta}{2e}$$

L is the diameter of the lens or aperture, and δ is the displacement of the image plane from the focal point.

Fundamentals of Computer Vision

- Decreasing the aperture or lens size to a point, such as in a pinhole camera, results in the blur circle's radius approaching zero.
- This optical effect increases the depth of field, eventually leading to all objects being in focus.
- However, reducing the aperture opening allows less light to form the image on the image plane.
- Pinhole cameras are practical only in bright circumstances due to the limited light intake resulting from the smaller aperture size.
- Camera optics can be tailored to suit the depth range required for a specific application.
- A zoom lens with a substantial focal length allows for range resolution over significant distances.
- However, this comes at the cost of a reduced field of view.
- Conversely, using a large lens diameter along with a fast shutter speed results in larger and more detectable blur circles.

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Mobile and Autonomous Robots

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Perception in Autonomous Robots

Robot vision basics - UGVs / UAVs/AUVs

Robot vision basics - UGVs

- An unmanned ground vehicle (UGV) is a vehicle that operates while in contact with the ground and without an onboard human presence.
- UGVs can be used for many applications where it may be inconvenient, dangerous, or impossible to have a human operator present.
- Generally, the vehicle will have a set of sensors to observe the environment and will either autonomously make decisions about its behavior or pass the information to a human operator at a different location who will control the vehicle through teleoperation.
- The UGV system generally consists of four parts such as vehicle control system, navigation system, and obstacle detecting system and traffic signal monitoring and detection system. leading to more robust decision-making.
- They are used in diverse fields, including military operations, search and rescue, agriculture, mining, logistics, and inspection.

Fundamentals of Computer Vision

Robot vision basics - UGVs

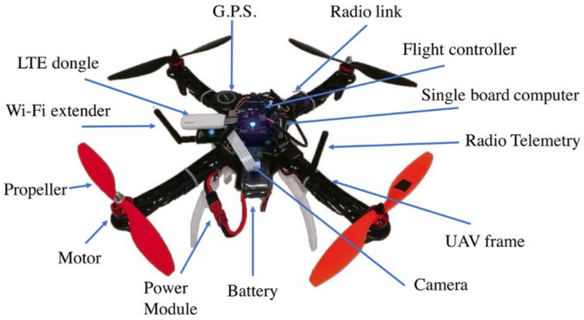
- Robotic vision sensing plays a vital role in the navigation, perception, and interaction of Unmanned Ground Vehicles (UGVs) with their environment.
- Acting as the "eyes" of these robots, it encompasses various technologies like cameras (RGB, thermal, depth), LiDAR, and ultrasonic sensors.
- Each technology provides unique capabilities: cameras offer visual information for object recognition, LiDAR creates detailed 3D maps for obstacle detection, and ultrasonic sensors excel in close-range obstacle avoidance.
- They assist with object manipulation and interaction, allowing robots to pick and place objects, perform complex tasks like grasping or assembly, and even interact with their surroundings safely.

Challenges:

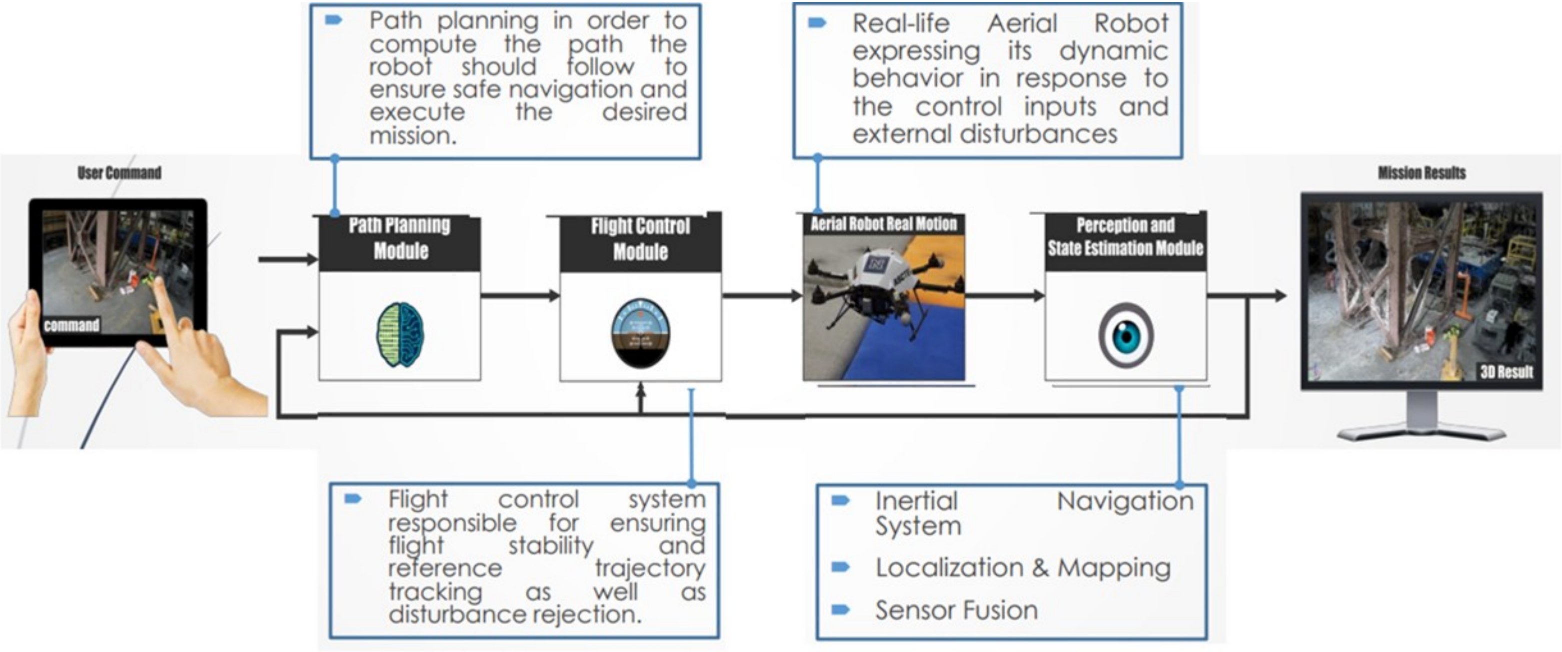
- Environmental factors like lighting, weather, and occlusions can significantly impact sensor performance.
- Moreover, computational limitations and power consumption require careful optimization. To overcome these hurdles, a multi-sensory approach is crucial.
- Fusing data from multiple sensors provides a more comprehensive understanding of the environment, leading to more robust decision-making.

Robot vision basics - UAVs

- The term aerial robotics is often attributed to Robert Michelson, as a way to capture a new class of highly intelligent, small flying machines·
- An aerial robot is a system capable of sustained flight with no direct human control and able to perform a specific task.
- The range of systems and activities covered under the label aerial robotics could extend much further
- The UAVs may operate either under remote control or autonomously with on-board computers.
- A typical UAV or drone consists of a frame, propulsion system (motors and propellers), flight control system, communication system (remote control and receiver), power system (batteries), and optional components like cameras and sensors.



Robot vision basics - UAVs



Block diagram of the main loops running at every aerial robot

Fundamentals of Computer Vision

The block diagram illustrates the main operational loops in an **aerial robot system**, detailing the different modules responsible for its navigation, control, perception, and mission execution. Here's a breakdown of the components:

- **User Command**
  - A user provides commands via a control interface (such as a tablet or computer).
  - These commands define the mission objectives, such as waypoints for navigation.
- **Path Planning Module**
  - Computes the optimal path for the aerial robot to follow.
  - Ensures safe navigation while considering obstacles and mission constraints.
- **Flight Control Module**
  - Ensures **flight stability** and **trajectory tracking**.
  - Compensates for external disturbances to maintain control.
- **Aerial Robot Real Motion**
  - The robot executes the computed path and reacts dynamically to control inputs and environmental disturbances.
  - The **real-life movement** of the aerial robot is influenced by external factors such as wind, obstacles, and sensor feedback.
- **Perception and State Estimation Module**
  - Uses various sensors to perceive the environment and estimate the robot's position.
  - Includes **Inertial Navigation System (INS), Localization & Mapping, and Sensor Fusion** to improve accuracy.
- **Mission Results**
  - The final output of the mission (e.g., images, 3D maps, or object detection results) is displayed on a screen.
  - This output helps in evaluating the mission's success and can be used for further analysis.

Fundamentals of Computer Vision

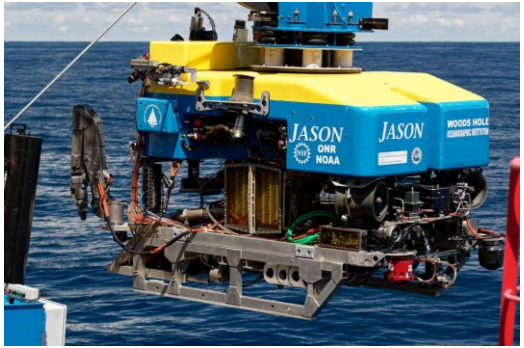
- Cameras, ranging from standard RGB to specialized thermal and multispectral options, provide robots with visual data for various tasks.
- Algorithms process this data to extract meaningful information, enabling tasks like object recognition, terrain mapping, and obstacle detection.
- This allows robots to navigate autonomously, avoid collisions, and even perform complex tasks like package delivery or infrastructure inspection.

  RGB cameras: Provide colour information for object recognition and scene understanding.
- Thermal cameras: Detect heat signatures, useful for low-light conditions or identifying people in dense foliage.
- Multispectral cameras: Capture data beyond the visible spectrum, aiding in agriculture, environmental monitoring, and resource exploration.
- However, challenges remain, including limited sensor resolution in low-light conditions, computational constraints for real-time processing on board, and potential safety concerns in congested environments

Robot vision basics - AUVs

- **Autonomous Underwater Vehicles (AUVs) are free- swimming unoccupied underwater vehicles that can overcome all limitations of Remotely Operated Vehicles (ROVs)**

- Such vehicles carry their own energy supplies (presently batteries, perhaps fuel cells in the future) and
- **communicate only through acoustics and perhaps optical links in the near future.**
- Limited communications require these vehicles to operate independently of continuous human control, in many cases the vehicles operate completely autonomously.
- AUVs are currently used for scientific survey tasks, oceanographic sampling, underwater archaeology and under-ice survey.



Robot vision basics - AUVs

- Underwater vehicles are equipped with a sensor system devoted to enabling motion control as well as accomplishing the specific mission it has been commanded to complete.
  - In the latter case, sensors developed for chemical/biological measurements or mapping may be installed
- AUVs need to operate underwater most of the time;
- One of the major problems with underwater robotics is in the localization task due to the absence of a single, proprioceptive sensor to measure the vehicle position.
- **The global position system (GPS) cannot be used underwater**
- Redundant multisensor systems are commonly combined using state estimation or sensor fusion techniques to provide fault detection and tolerance capability to the vehicle

Robot vision basics - AUVs

The sensors that can be found on an underwater vehicle are:

- **Compass:** A gyrocompass can provide an estimate of geodetic (true) north accurate to a fraction of a degree.
  - Magnetic compasses can provide estimates of magnetic north with an accuracy of less than 1◦ if carefully calibrated to compensate for magnetic disturbances from the vehicle itself
- **Inertial measurement unit (IMU):** An IMU provides information about the vehicle’s linear acceleration and angular velocity.
  - These measurements are combined to form estimates of the vehicle’s attitude including an estimate of geodetic north from the most complex units.
- **Depth sensor:** Measuring the water pressure gives the vehicle’s depth.
  - At depths beyond a few hundred meters, the equation of state of seawater must be invoked to produce an accurate depth estimate based on the ambient pressure

General

Robot vision basics - AUVs

- **Altitude and forward-looking sonar:** These are used to detect the presence of obstacles and distance from the seafloor

- **Global positioning system (GPS):** This is used to localize the vehicle while on the surface to initialize or reduce drift of estimates from an IMU/DVL combination. GPS only works at the surface.

- **Acoustic positioning:** A variety of schemes exist for determining vehicle position using acoustics

- **Vision systems:** Cameras can be used to obtain estimates of relative, and in some cases absolute, motion using a type of simultaneous localization and mapping (SLAM) algorithm and used to perform tasks such as visual tracking of pipelines, station keeping, visual servoing or image mosaicking.

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Mobile and Autonomous Robots

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Perception

Fundamentals of Image Processing

Feature extraction

Mobile and Autonomous Robots

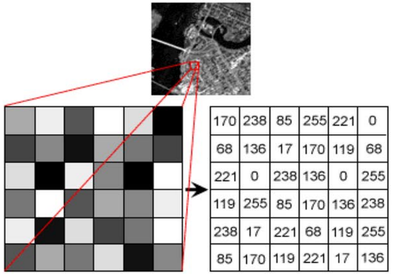
Introduction to Image Processing

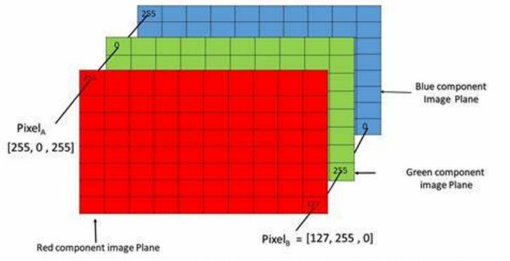
Image Processing

- Image Processing involves manipulation and analysis of images using algorithms
- Used to enhance image quality or extract meaningful information
- Forms the foundation of Computer Vision and AI systems

Digital Image Representation

- An image is represented as a matrix of pixels
- Each pixel has:
  - **Location**: (row, column)
  - **Intensity value**
- Types:
  - **Binary image: 0 or 1**
  - **Grayscale image: typically 0–255 (8-bit)**
  - **Color image: RGB (Red, Green, Blue channels)**





Pixel of an RGB image are formed from the corresponding pixel of the three component images

Mobile and Autonomous Robots

Introduction to Image Processing

Image Acquisition

- Steps involved:
  - **Sensing** – Camera, CCD/CMOS sensor
  - **Sampling** – Converting continuous image to discrete pixels
  - **Quantization** – Assigning discrete intensity levels
- Key idea:
  - **Higher sampling** → better spatial resolution
  - **Higher quantization** → better intensity resolution

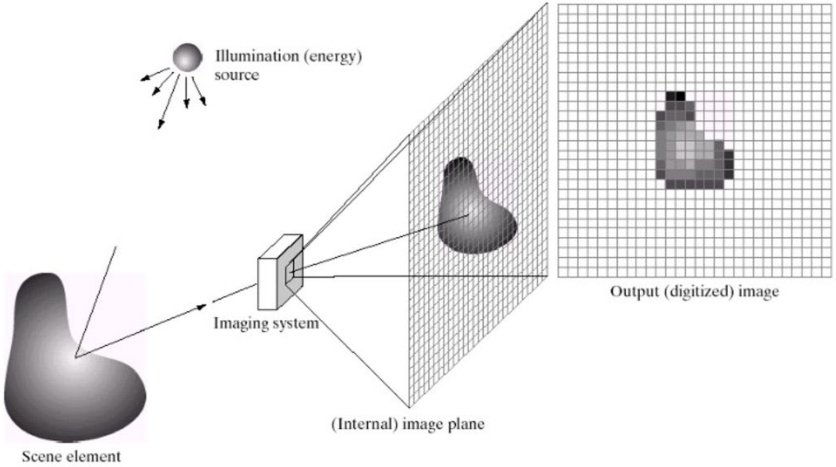


Image Processing Pipeline

- Image acquisition
- Preprocessing (noise removal, enhancement)
- Segmentation
- Feature extraction
- Interpretation / Decision making

Feature Extraction

- Autonomous mobile robots rely on sensor measurements to determine their relationship to the environment.
- Sensors introduce uncertainty and error into measurements, requiring strategies to mitigate these effects.
- Two main strategies for using uncertain sensor input are:
    - Using raw sensor measurements directly to inform robot behavior.
    - Extracting higher-level perceptual information from sensor readings before informing robot behavior.

Feature Extraction

- Feature extraction is the process of extracting higher-level percepts from sensor readings.
- Mobile robots vary in their use of feature extraction and scene interpretation depending on specific functionalities.
- For immediate obstacle detection, robots may directly use raw sensor readings to trigger emergency stops.

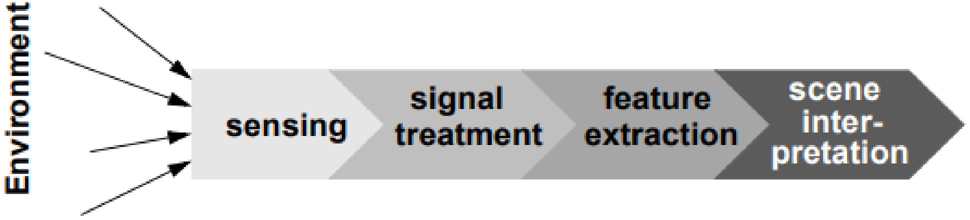


Figure 4.63
The perceptual pipeline: from sensor readings to knowledge models.

Feature extraction

- For local obstacle avoidance, raw sensor readings may be combined in an occupancy grid model for smooth avoidance.
- For map-building and precise navigation, sensor values may undergo feature extraction and scene interpretation to minimize individual sensor uncertainty.
- More sophisticated perceptual tasks require greater reliance on feature extraction and scene interpretation.

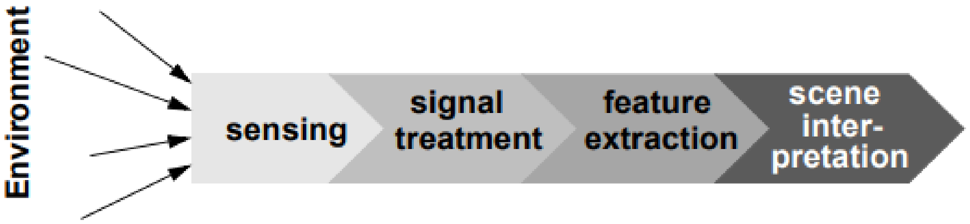


Figure 4.63
The perceptual pipeline: from sensor readings to knowledge models.

Feature Definition

- Features are **recognizable structures** in the environment that can be **mathematically described**.
- They can be categorized into **low-level features** (geometric primitives) and **high-level features** (objects).
- Low-level features include **lines, points, corners, blobs, circles, or polygons.**
- High-level features include **objects like doors, tables, or trash cans**.
- Raw sensor data provide a large volume of data but with low distinctiveness, while low-level features abstract raw data, increasing distinctiveness but reducing data volume.

Feature Definition

- High-level features provide maximum abstraction from raw data, further reducing data volume while providing highly distinctive features.
- Features must have some spatial locality, but their geometric extent can vary widely.
- In mobile robotics, features are crucial for creating environmental models, enabling compact and robust descriptions of the environment for map-building and localization.
- Choosing the appropriate features for a mobile robot involves considering factors such as the environment, task requirements, and sensor capabilities.

Target Environment

- Geometric features must be easily detectable in the actual environment to be useful.
- Line features are highly useful in environments with abundant straight wall segments, such as office buildings.
- Conversely, line features are virtually useless in environments like Mars where such structures are scarce.
- Point features, such as corners and blobs, are more likely to be found in any textured environment.
- For instance, NASA's Mars exploration rovers, Spirit and Opportunity, utilized corner features for visual odometry, leveraging the textured environment of Mars.

Feature extraction

Available sensors

- The choice of features for a robot depends on its specific sensors and the uncertainty associated with them.
- Robots equipped with laser rangefinders are well-suited for using geometrically detailed features like corner features.
- Laser scanners offer high-quality angular and depth resolution, making them ideal for extracting precise geometric features.
- In contrast, robots equipped with sonar sensors may not have the necessary capabilities for corner feature extraction.
- The appropriateness of various features is directly impacted by the sensors available to the robot for environmental perception.

Feature extraction

Computational power

- Visual feature extraction can incur a significant computational cost, especially in robots where the processing of vision sensor data is handled by the main processor.
- The process of extracting visual features involves analyzing raw image data to identify and describe recognizable structures or patterns.
- This computational overhead arises from the need to process large volumes of image data and perform complex algorithms to extract meaningful features.
- Robots relying on vision sensors for perception tasks may face challenges in real-time processing due to the computational demands of feature extraction.
- Efficient algorithms and hardware acceleration techniques may be employed to mitigate the computational burden associated with visual feature extraction.
- Balancing computational resources with the need for accurate feature extraction is crucial for optimizing the performance of vision-based robotic systems.

Feature extraction

Environment representation

- Feature extraction is a crucial step towards scene interpretation, ensuring that the extracted features align with the representation used for the environmental model.
- Different representations of the environment, such as geometric and topological models, require distinct types of features for effective modeling.
- Line and corner features hold more relevance in geometric environmental models compared to non-geometric visual features, which are more suitable for topological models.
- Vision interpretation poses significant challenges, with decades of research dedicated to developing algorithms for understanding scenes based on 2D images.

Feature extraction

Environment representation

- Commercially available vision sensors for mobile robots, such as vision ranging and color-tracking sensors, have emerged due to relatively focused challenges and straightforward algorithms.
- Point feature extraction techniques relevant to mobile robotics must meet two key requirements: real-time operation and robustness to real-world conditions.
- Vision interpretation primarily involves the challenge of reducing information, as sensors like CCD cameras produce vast amounts of data containing both relevant and irrelevant information.
- The problem of visual feature extraction involves removing irrelevant information from images to ensure that the remaining information accurately describes specific features in the environment.

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HARRIS CORNER DETECTION

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Harris Corner Detection

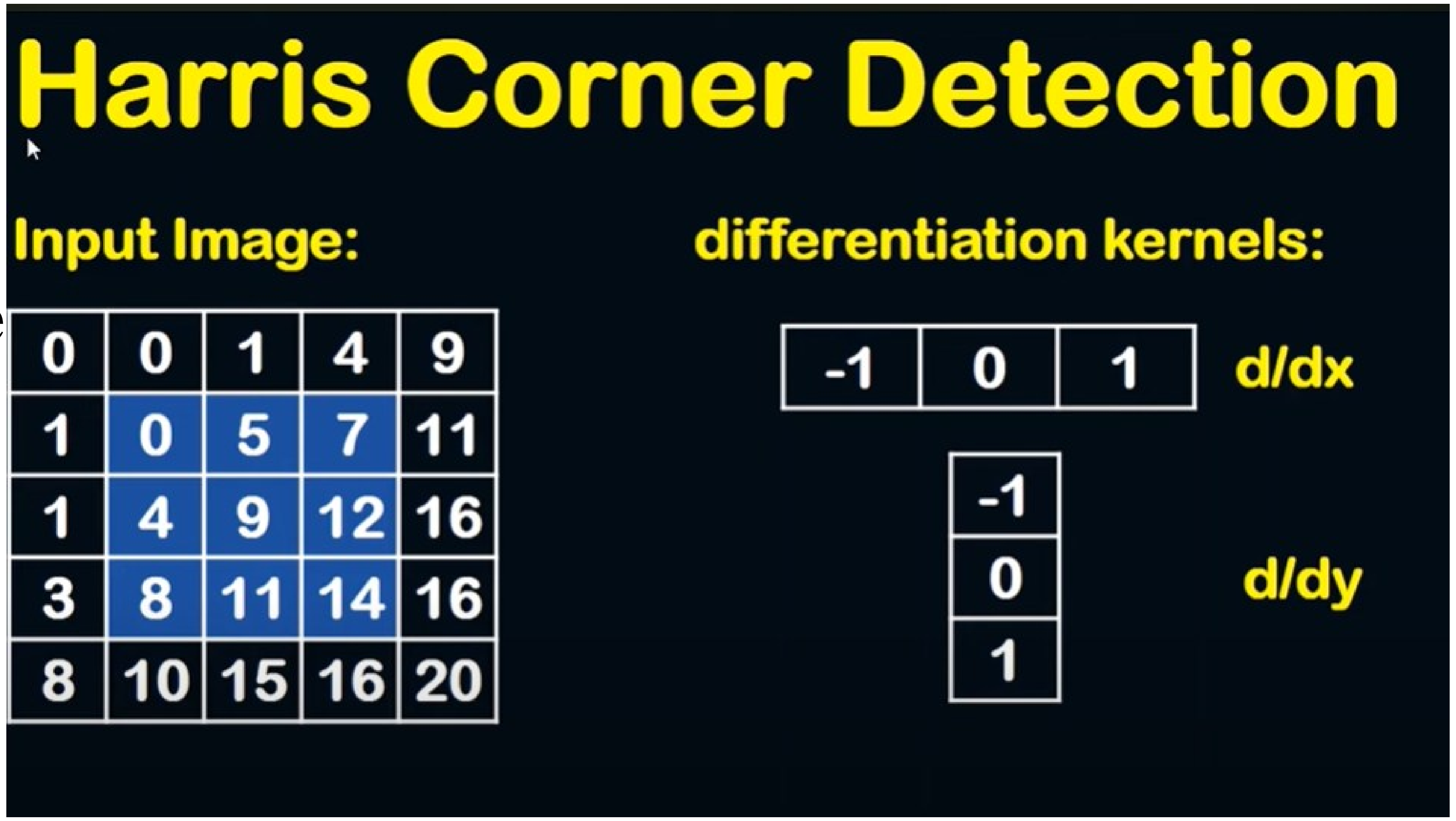
Introduction to Harris Corner Detection:

- Harris Corner Detection is a feature detection algorithm used in computer vision to identify corners in an image. Corners are points where the intensity of the image changes significantly in multiple directions. This method is widely used in object tracking, image stitching, and motion detection. The algorithm calculates changes in intensity using derivatives and determines whether a region is a corner, an edge, or a flat area based on the Harris Corner Response function.

Harris Corner Detection

Input Image and Selected Region

- **The input image is given, with a highlighted region where we will apply Harris Corner Detection.**
- **The goal is to compute the Harris Matrix for this region.**
- **The differentiation kernels for X and Y directions are provided.**

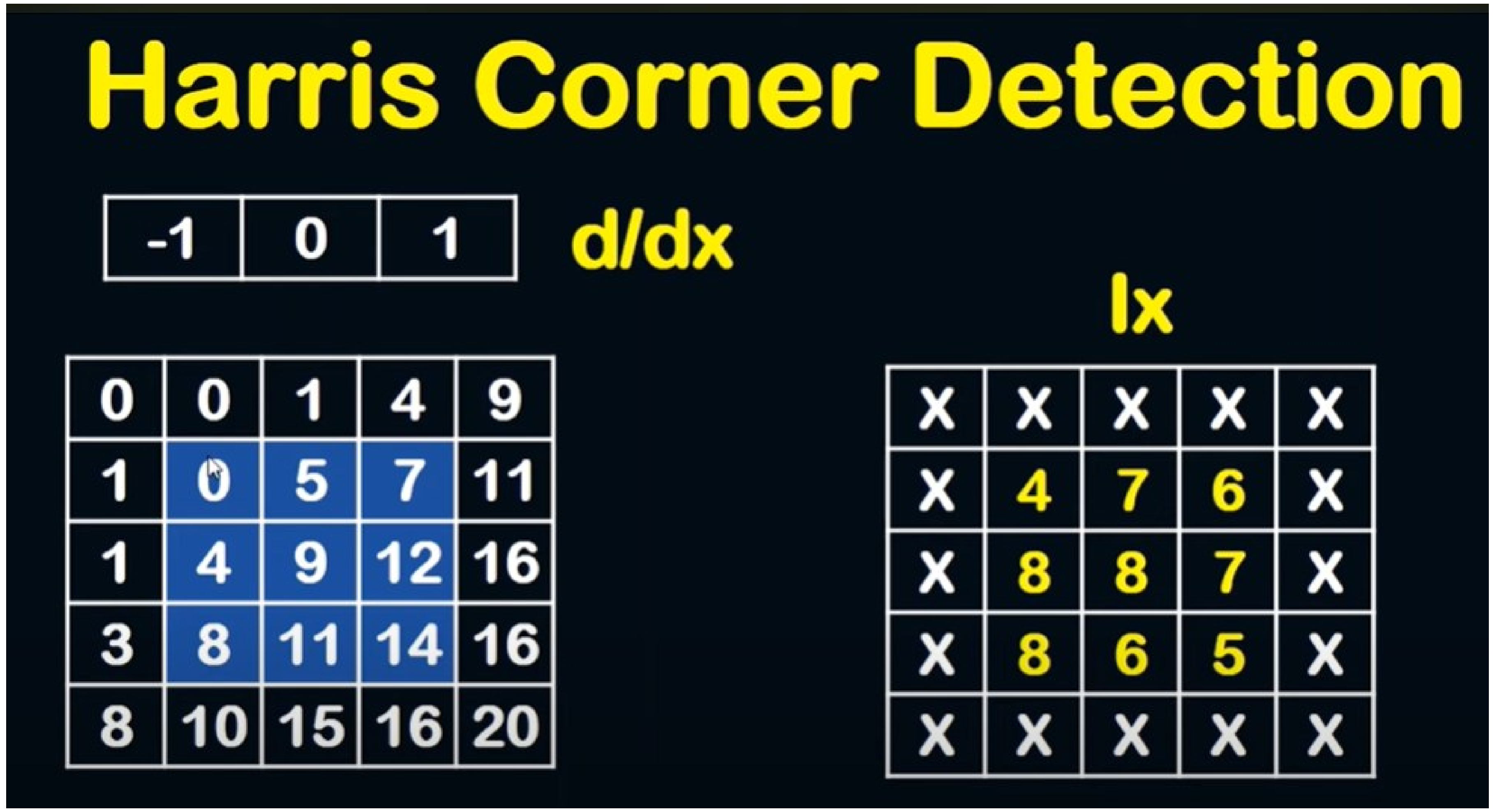


Kernels Used:
X-Direction: [-1, 0, 1]
Y-Direction: [[-1], [0], [1]]

Harris Corner Detection

Computing the X-Derivative

- Apply the differentiation kernel in the X direction.
- Place the center of the kernel on the pixel.
- Compute the convolution by multiplying and summing the values.



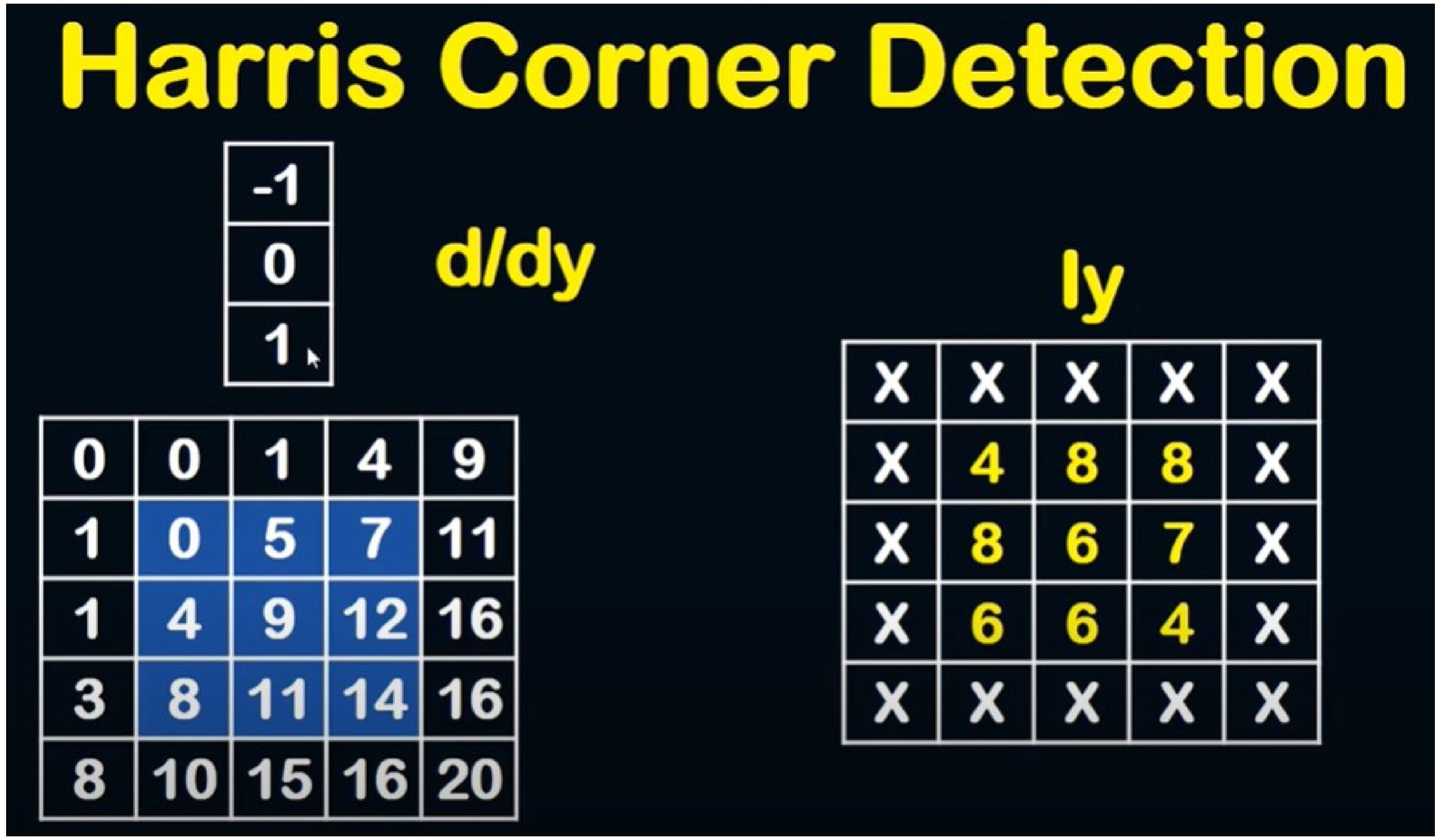
Example of X-Derivative Computation

- For a given pixel:
  - (-1 * Left Pixel) + (0 * Center Pixel) + (1 * Right Pixel)
- Example computation for the 0 pixel of the highlighted region:
  - (-1 * 1) + (0 * 0) + (1 * 5) = -1 + 0 + 5 = 4
- Repeat for all pixels in the region.

Harris Corner Detection

Computing the Y-Derivative

- Apply the differentiation kernel in the Y direction.
- Similar to X-Derivative, but the kernel moves vertically.



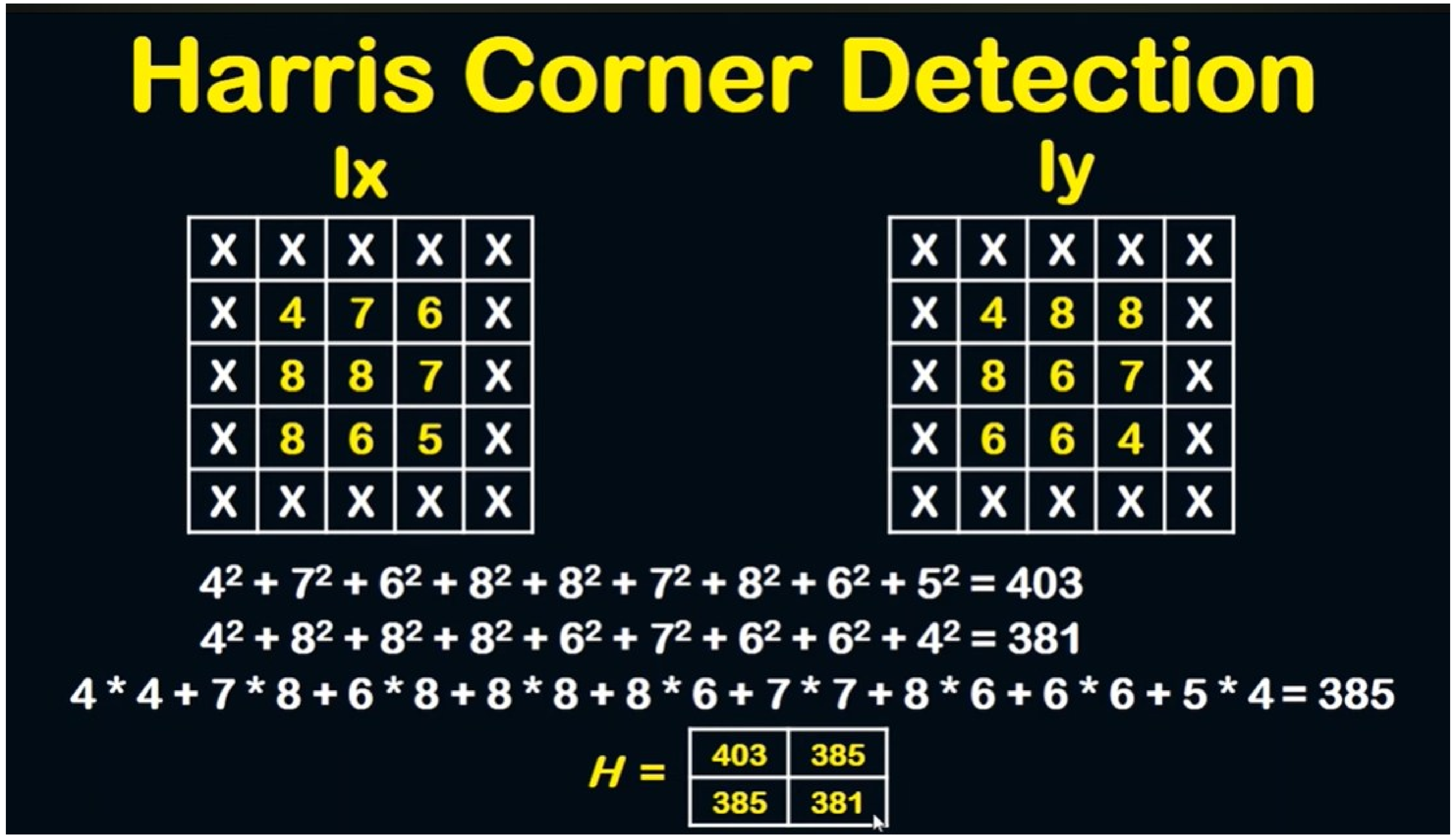
- For a given pixel:
    - (-1 * Top Pixel) + (0 * Center Pixel) + (1 * Bottom Pixel)
- Example computation for the 0 pixel of the highlighted region:

  (-1*0) + (0*0) + (1*4) = 4
- Compute for all pixels in the selected region.

Harris Corner Detection

Computing the Harris Matrix Components

- Using the computed X and Y derivatives:
- Compute the squared sum of X-derivatives.
- Compute the squared sum of Y-derivatives.
- Compute the product sum of X and Y derivatives.
- Compute:
  - Ix^2, Iy^2, Ix * Iy
- Construct the 2x2 Harris Matrix:
- H = [Σ(Ix²) Σ(IxIy)]
  [Σ(IxIy) Σ(Iy²)]



Harris Corner Detection

Computing Harris Response (Corner Score)

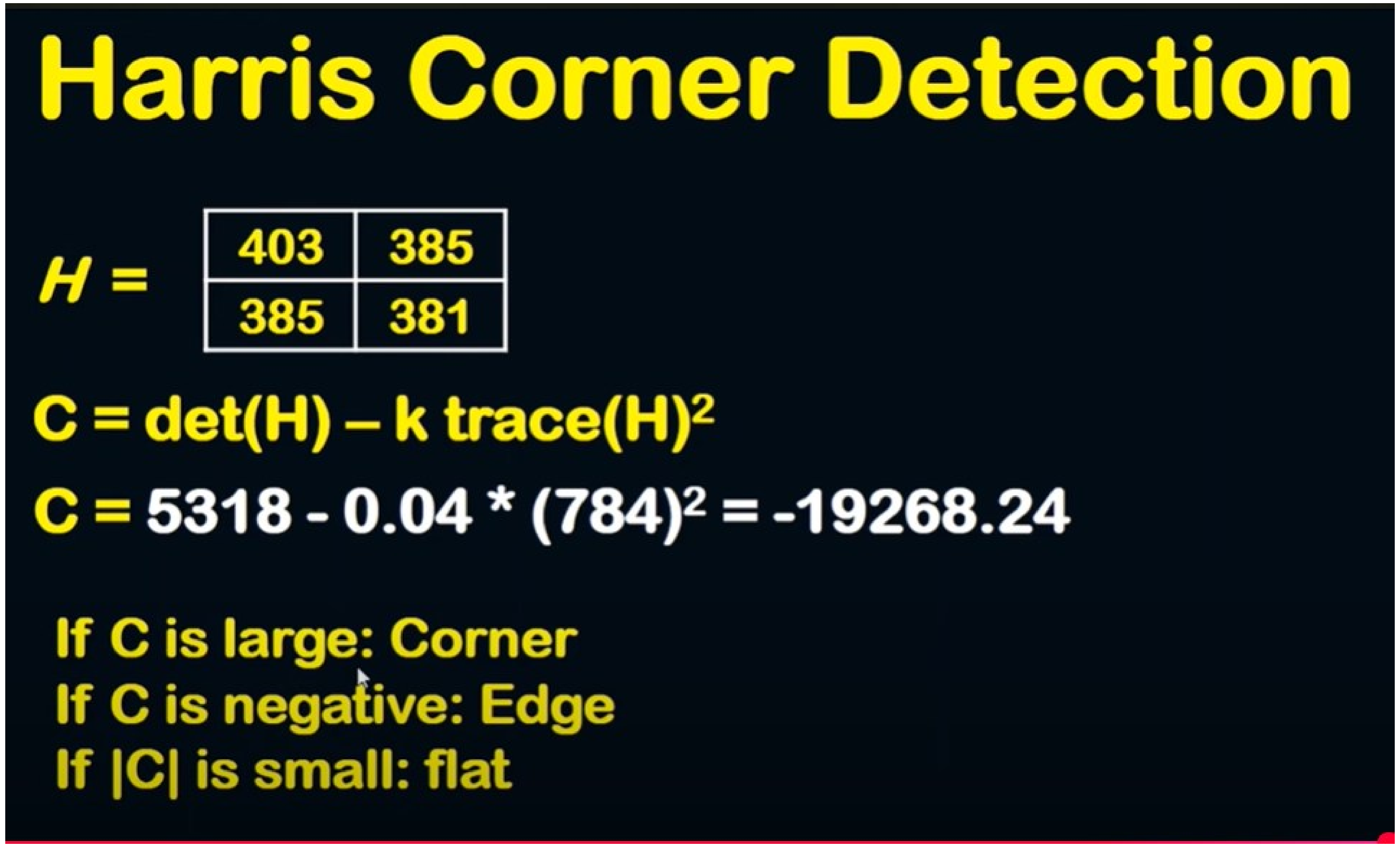
- **Formula:**
  - $C = \det(H) - k * (\text{trace}(H))^2$

where,

- $\det(H) = (H_{xx} * H_{yy}) - (H_{xy} * H_{xy})$
- $\text{trace}(H) = H_{xx} + H_{yy}$
- k is a constant (typically 0.04)

Interpreting the Harris Corner Score

- If C is a large positive value → Corner
- If C is negative → Edge
- If |C| is small → Flat region.



Harris Corner Detection

Applications of Harris Corner Detection

- Feature extraction in object recognition.
- Keypoint detection in tracking systems.
- Used in Structure from Motion (SfM) and Simultaneous Localization and Mapping (SLAM).

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Perception

Image Feature Extraction

Image Feature Extraction: Interest Point Detectors

- Local features, also known as interest points, interest regions, or keypoints, are image patterns that differ from their immediate surroundings in terms of intensity, color, and texture.
- They can manifest as small image patches, edges, or points, such as corners resulting from line intersections.
- Local features can be categorized based on their semantic content:
- The first category includes features with a semantic interpretation, such as edges corresponding to road lanes or blobs representing blood cells in medical images.

Image Feature Extraction: Interest Point Detectors

- The second category comprises features without a semantic interpretation; their accurate and robust location over time is essential for applications like feature tracking, camera calibration, 3D reconstruction, and image mosaicing.
- The third category consists of features that lack individual semantic interpretation but contribute to scene or object recognition when considered collectively. Scene recognition, object recognition, texture analysis, scene classification, and image retrieval are common application domains.
- Visual-word-based place recognition, which relies on the matching of features between observed scenes and query images, is an example of utilizing features to recognize scenes or objects collectively.

Properties of the ideal feature detector

- **Repeatability:** A good feature detector should ensure that a high percentage of features detected in one image can be reliably rediscovered in another image of the same scene, even under different viewing angles and illumination conditions.
- **Distinctiveness:** Features should be highly distinguishable from each other, allowing for accurate matching between images. The surrounding patch of a feature point should carry distinctive information.
- **Localization accuracy**: Detected features should be precisely localized in both image position and scale, especially crucial for tasks like camera calibration, 3D reconstruction, and panorama stitching.
- **Quantity of features:** The number of detected features should be appropriate for the application, with tasks like object recognition and 3D reconstruction benefitting from a larger number of features.

Properties of the ideal feature detector

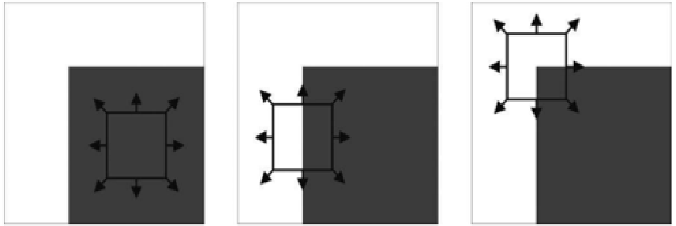
- **Invariance:** Features should exhibit invariance to changes in camera viewpoint, environmental illumination, and scale (e.g., zoom or camera translation), achievable through mathematical transformations.
- **Computational efficiency:** Efficient detection and matching of features are essential, particularly for real-time applications like robotics and large-scale image processing projects such as Google Images. Computational time may vary based on the desired level of invariance.
- **Robustness:** Detected features should be robust to various factors like image noise, compression artifacts, blur, and deviations from mathematical models used for achieving invariance.

Corner Detectors

A corner in an image can be defined as the intersection of two or more edges. Corners are features with high repeatability.

The basic concept of corner detection

An intuitive explanation of Moravec's corner detection algorithm is given in the below figure.



(a) Flat region (b) Edge (c) Corner

Interest Point Detectors

- One could recognize a corner by looking through a small window centered on the pixel.
- Moravec observed that in a "flat" region of an image (uniform intensity), adjacent windows will look similar.
- Along an edge, windows perpendicular to the edge will look different, while those parallel to the edge will result in only a small change.
- At a corner, none of the adjacent windows will appear similar.
- Moravec utilized the Sum of Squared Differences (SSD) as a measure of similarity between two patches.
- A low SSD value indicates greater similarity between patches.
- Local maximization of SSD indicates the presence of a corner.

The Harris corner detector

Harris and Stephens improved Moravec's corner detector by considering the partial derivatives of the SSD instead of using shifted windows.

Let I be a grayscale image. Consider taking an image patch centered on (u, v) and shifting it by (x, y). The Sum of Squared Differences SSD between these two patches is given by:

$$SSD(x, y) = \sum_u \sum_v ((I(u, v)) - I(u + x, v + y))^2 .$$

$I(u + x, v + y)$ can be approximated by a first-order Taylor expansion. Let I_x and I_y be the partial derivatives of I, such that

$$I(u + x, v + y) \approx I(u, v) + I_x(u, v)x + I_y(u, v)y .$$

Interest Point Detectors

This produces the approximation

$$SSD(x, y) \approx \sum \sum (I_x(u, v)x + I_y(u, v)y)^2 ,$$

which can be written in matrix form:

$$SSD(x, y) \approx \begin{bmatrix} x & y \end{bmatrix} M \begin{bmatrix} x \\ y \end{bmatrix} ,$$

where *M* is the second moment matrix.

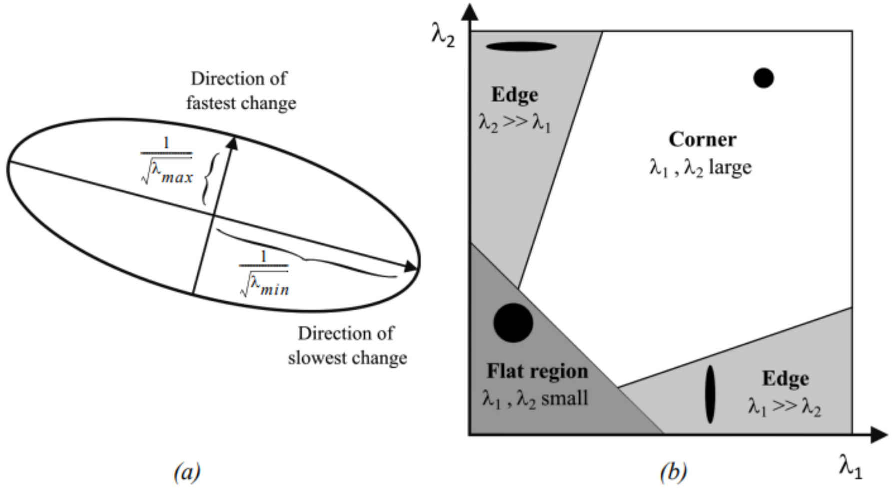


Figure 4.67 (a) This ellipse is built from the second moment matrix and visualizes the directions of fastest and lowest intensity change. (b) The classification of corner and edges according to Harris and Stephens.

$$M = \sum_u \sum_v \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} \sum\sum I_x^2 & \sum\sum I_x I_y \\ \sum\sum I_x I_y & \sum\sum I_y^2 \end{bmatrix}.$$

And since M is symmetric, we can rewrite M as

$$M = R^{-1} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} R,$$

where λ_1 and λ_2 are the eigenvalues of M.

The Harris detector analyses the eigenvalues of to decide if we are in presence of a corner or not.

The axis lengths of this ellipse are determined by the eigenvalues of *M* and the orientation is determined by *R* .

Interest Point Detectors

Based on the magnitudes of the eigenvalues, the following inferences can be made based on this argument:

- If both λ_1 and λ_2 are small, SSD is almost constant in all directions (i.e., we are in presence of a flat region).
- If either $\lambda_1 \gg \lambda_2$ or $\lambda_2 \gg \lambda_1$, we are in presence of an edge: SSD has a large variation only in one direction, which is the one perpendicular to the edge.
- If both λ_1 and λ_2 are large, SSD has large variations in all directions and then we are in presence of a corner.

Interest Point Detectors

Because the calculation of the eigenvalues is computationally expensive, Harris and Stephens suggested the use of the following "cornerness function" instead:

$$C = \lambda_1\lambda_2 - \kappa(\lambda_1 + \lambda_2)^2 = det(M) - \kappa \cdot trace^2(M),$$

where κ is a tunable sensitivity parameter. This way, instead of computing the eigenvalues of M, we just need to evaluate the determinant and trace of M. The value of κ has to be determined empirically. In the literature, values are often reported in the range 0.04–0.15.

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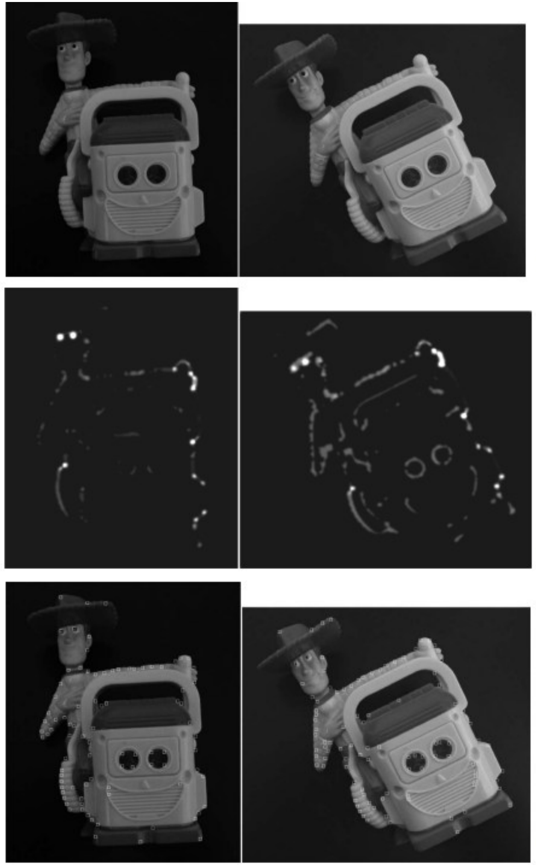
Interest Point Detectors

- This way, instead of computing the eigenvalues of M, we just need to evaluate the determinant and trace of M.
- The value of k has to be determined empirically.
- Values are often reported in the range 0.04–0.15.
- The last step of the Harris corner detector consists in extracting the local maxima of the cornerness function, using nonmaxima suppression.
- Finally only the local maxima which are above a given threshold are retained.



(a)

(b)

(c)

Figure 4.68 Extraction of the Harris corners. (a) Original image. (b) Cornerness function. (c) Harris corners are identified as local maxima of the cornerness function (only local maxima larger than a given threshold are retained). Two images of the same object are shown, which differ by illumination and orientation.

Invariance to photometric and geometric changes

Image transformations can affect the geometric or the photometric properties of an image. Consolidated models of image transformations are the following:

- Geometric changes:
  - 2D Rotation
  - Scale (uniform rescaling)
  - Affine
- Photometric changes
  - Affine intensity

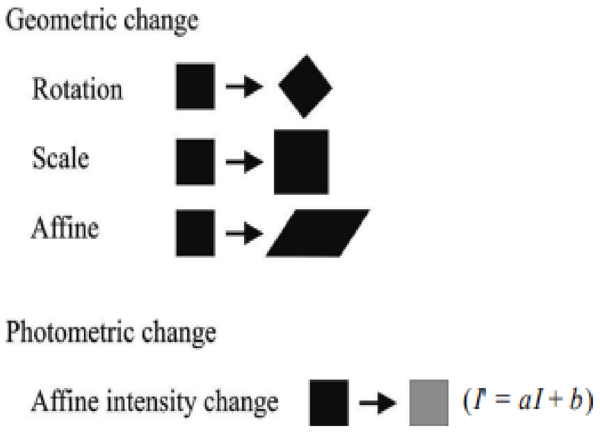


Figure 4.69 Models of image changes.

Interest Point Detectors

- The Harris detector is invariant to 2D image rotations because the eigenvalues of the second moment matrix remain unchanged under pure rotation.
- Affine intensity changes do not affect the position of local maxima of the cornerness function in the Harris detector, although the eigenvalues and cornerness function may be rescaled.
- However, the Harris detector is not invariant to geometric affine transformations or scale changes.
- Geometric affine transformations distort the neighborhood of the feature, affecting its curvature and classification.
- Scale changes also impact the classification of features, with corners possibly being classified as edges at higher scales and corners at smaller scales.

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Interest Point Detectors

- A comparative study demonstrated that after rescaling an image by a factor of 2, only 20% of possible correspondences were detected again using the Harris detector.
- Despite not being invariant to scale changes, the Harris detector remains widely used and is found in popular computer vision libraries like OpenCV.
- Modifications to the original implementation have made the Harris detector invariant to scale and affine changes, and improvements in location accuracy can be achieved through subpixel precision approximation of the cornerness function.

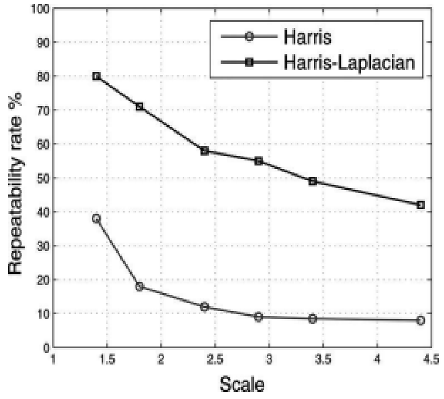

Figure 4.71 Repeatability rate of Harris detector. Comparison with Harris-Laplacian. This plot is the result of a comparative study presented in [216].

Interest Point Detectors

Scale-invariant detection

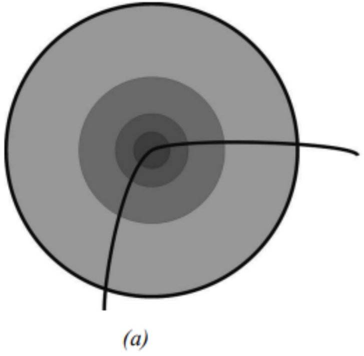

Figure 4.72 To achieve scale-invariant detection, the image is analyzed at different scales. This means, the Harris detector is applied several times on the image, each time with a different window size.

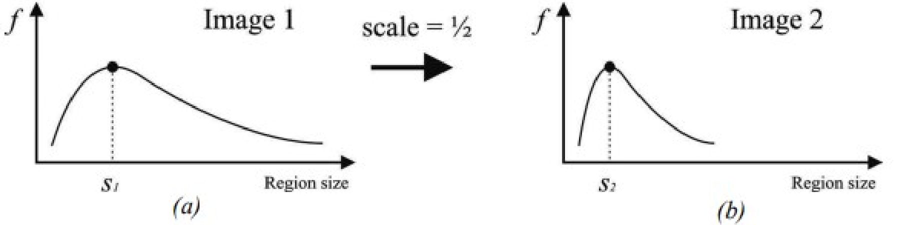

Figure 4.73 Average intensity as a function of the region size. (a) Original image. (b) Resized image.

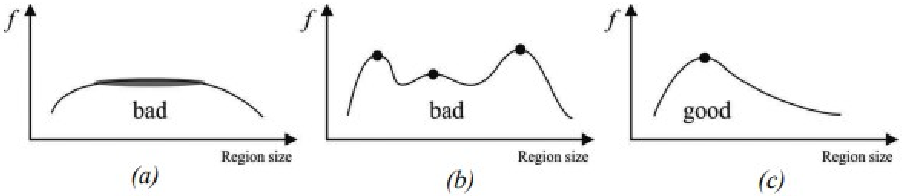

Figure 4.74 (a)-(b) are bad scale invariant functions. (c) is a good.

Interest Point Detectors

- If we look at figure 4.72 in the previous slide, we will notice that one way to detect the corner at higher scale is to use a multiscale Harris detector.
- This means that the same detector is applied several times on the image, each time with a different window (circle) size.
- An efficient implementation of multiscale detection uses the so called scale-space pyramid: instead of varying the window size of the feature detector, the idea is to generate upsampled or downsampled versions of the original image (i.e., pyramid).
- Using multiscale Harris detector, we can be sure that at some point the corner of image 4.72a will get detected.

Affine invariant detection

- Affine transformation is a good approximation of perspective distortion of locally planar patches under small viewpoint changes.
- The affine invariant Harris detector is known as Harris-Affine and was devised by Mikolajczyk and Schmid.

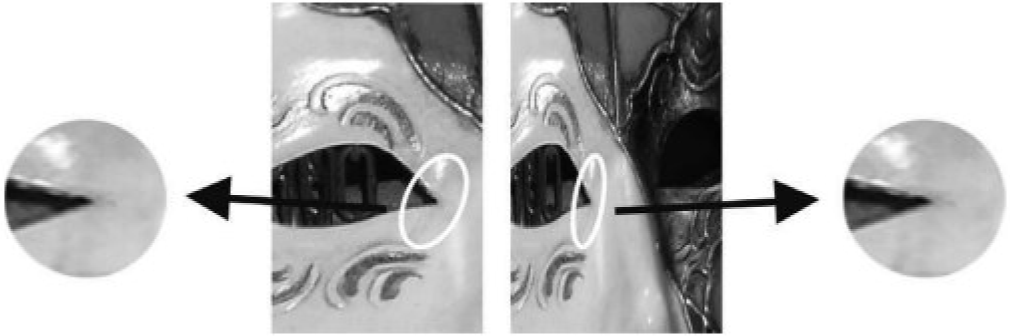

Figure 4.77 Computation of the affine invariant ellipse in two images related by an affine transformation.

Interest Point Detectors

How to detect the same features under affine transformation, which can be seen as a nonuniform rescaling?

The procedure consists in the following steps:

- First, the features are identified using the scale invariant Harris-Laplacian detector.
- Then, the second moment matrix is used to identify the two directions of slowest and fastest change of intensity around the feature.

$$M = \sum_u \sum_v \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} \sum\sum I_x^2 & \sum\sum I_x I_y \\ \sum\sum I_x I_y & \sum\sum I_y^2 \end{bmatrix}.$$

- Out of these two directions, an ellipse is computed to the same size as the scale computed with the LoG operator.
- The region inside the ellipse is normalized to a circular one.
- The initial detected ellipse and the resulting normalized circular shape are shown in figure 4.77

Other Corner detectors

- The Shi-Tomasi corner detector:
    - strongly based on the Harris corner detector.
- The SUSAN corner detector:
    - SUSAN stands for Smallest Univalue Segment Assimilating Nucleus and, besides being used for corner detection, it is also used for edge detection and noise suppression.
- The FAST corner detector:
    - FAST (Features from Accelerated Segment Test)

Interest Point Detectors

Table 4.2
Comparison of feature detectors: properties and performance.

	Corner detector	Blob detector	Rotation invariant	Scale invariant	Affine invariant	Repeatability	Localization accuracy	Robustness	Efficiency
Harris	x		x			+++	+++	++	++
Shi-Tomasi	x		x			+++	+++	++	++
Harris-Laplacian	x	x	x	x		+++	+++	++	+
Harris-Affine	x	x	x	x	x	+++	+++	++	++
SUSAN	x		x			++	++	++	+++
FAST	x		x			++	++	++	++++
SIFT		x	x	x	x	+++	++	+++	+
MSER		x	x	x	x	+++	+	+++	+++
SURF		x	x	x	x	++	++	++	++

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Moravec Corner Detection

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Moravec Corner Detection

Introduction:

- Topic: Corner Detection using Sum of Squared Differences (SSD)
- Objective: Identify corners in an image by comparing pixel neighborhoods

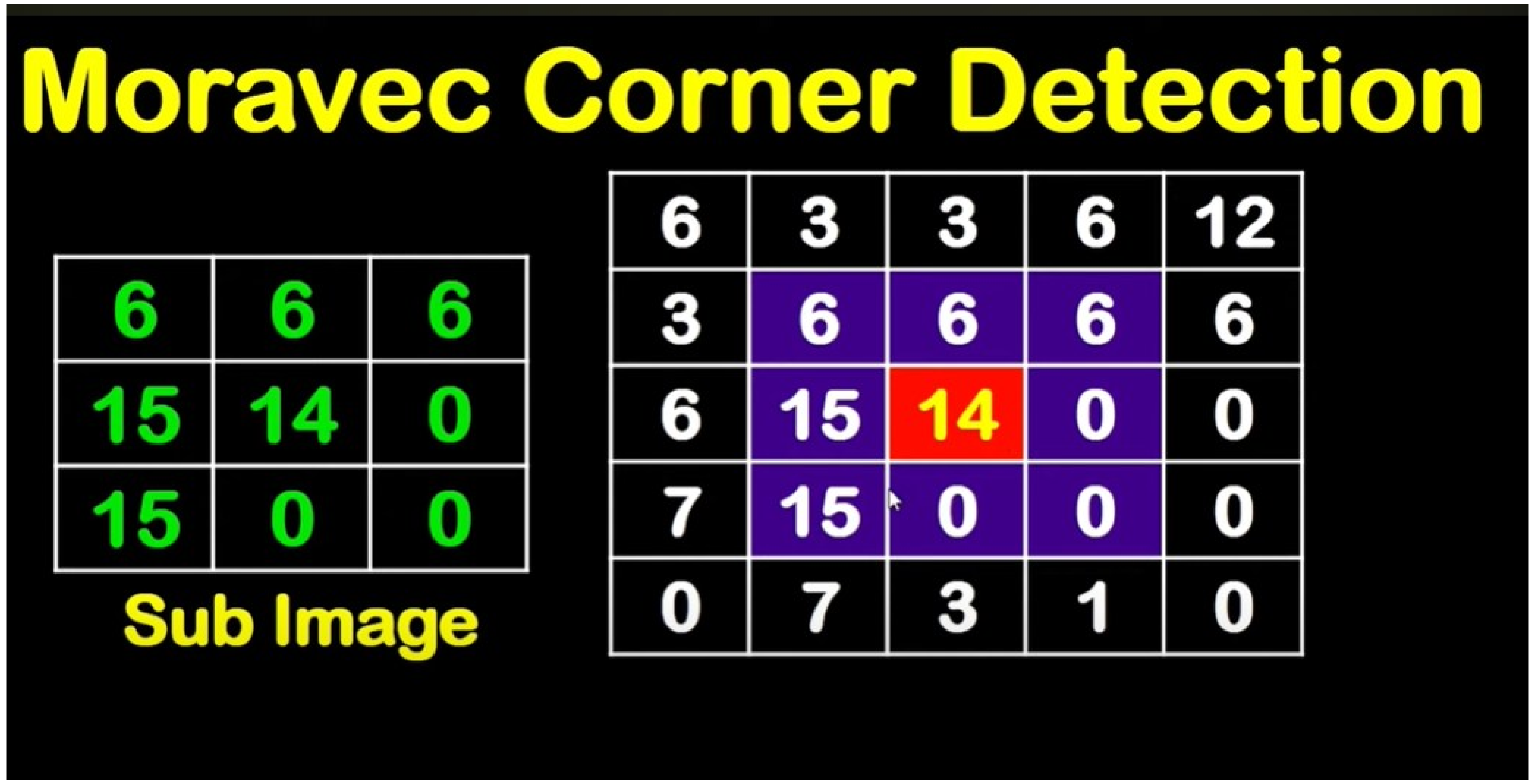
Importance:

- Used in feature detection for image processing
- Helps identify key points for object recognition and tracking

Method Overview:

- Define a reference 3×3 matrix around a target pixel
- Shift the matrix in different directions
- Compute the SSD (Sum of Squared Differences)
- Compare the SSD with a threshold to determine if the pixel is a corner

Moravec Corner Detection



Defining the Reference Matrix

- Select a target pixel (e.g., pixel value = 14)
- Draw a 3×3 neighborhood around the pixel
- This serves as the reference matrix

Example Reference Matrix:

[6   6   6]
[15 14 0] <- Target Pixel (14)
[15 0 0]

This reference matrix will be used to compute SSD

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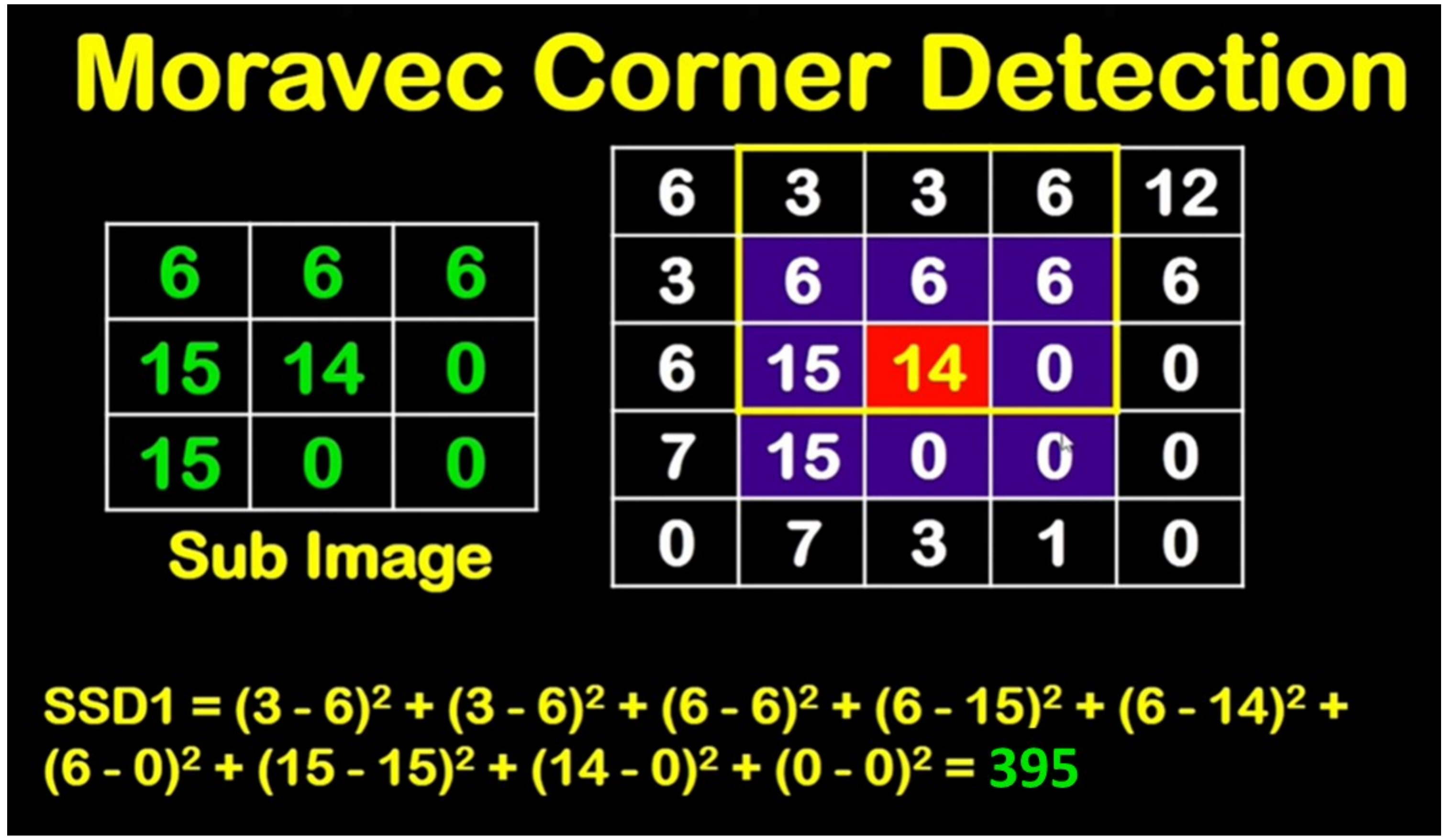
Shifting the Matrix

The reference matrix is shifted in eight directions:

1. Up (Move 1 pixel up)
2. Down (Move 1 pixel down)
3. Left (Move 1 pixel left)
4. Right (Move 1 pixel right)
5. Diagonal Up-Left (Move 1 pixel up & left)
6. Diagonal Up-Right (Move 1 pixel up & right)
7. Diagonal Down-Left (Move 1 pixel down & left)
8. Diagonal Down-Right (Move 1 pixel down & right)



Each shift creates a new 3×3 matrix for SSD calculation

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Computing SSD (Sum of Squared Differences)

SSD measures how different the shifted matrix is from the reference matrix

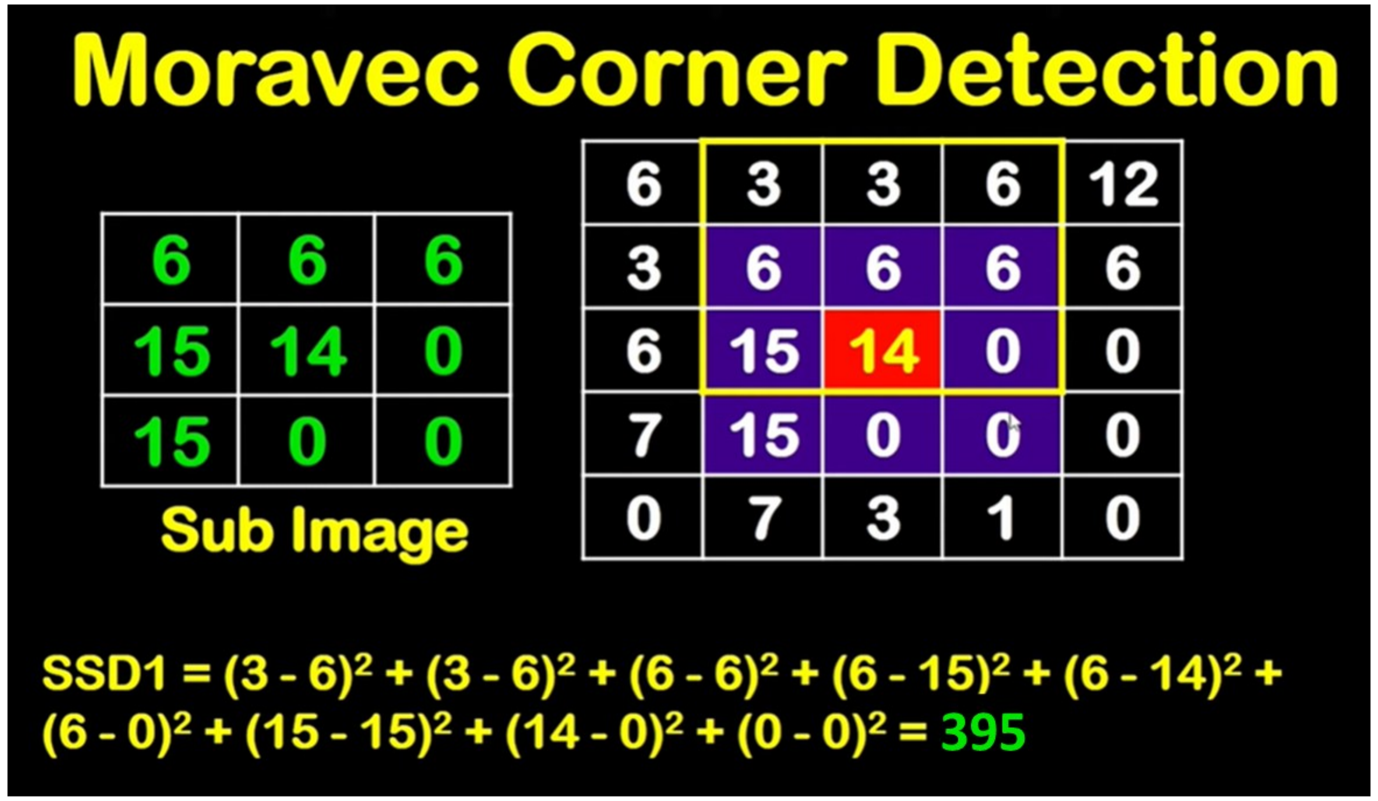
Formula:

$$SSD = \Sigma\ (\text{Reference Matrix} - \text{Shifted Matrix})^2$$

Example calculation for up shift:

$SSD1 = (3 - 6)^2 + (3 - 6)^2 + (6 - 6)^2 + (6 - 15)^2 + (6 - 14)^2 + (6 - 0)^2 + (15 - 15)^2 + (14 - 0)^2 + (0 - 0)^2 = 365$

This SSD value is stored for further comparison



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SSD Calculation for All Shifts

The SSD calculation is repeated for all eight shifts

- **The resulting SSD values are stored in a list**

- **SSD1 = 395**
- **SSD2 = 451**
- **SSD3 = 422**
- **SSD4 = 576**
- **SSD5 = 227 (Minimum SSD)**
- **SSD6 = 764**
- **SSD7 = 493**
- **SSD8 = 702**

Moravec Corner Detection

Selecting the Minimum SSD

- The smallest SSD value is chosen

In this example:

- Minimum SSD = 227 (SSD5)

This minimum SSD represents the most similar neighborhood

Thresholding for Corner Detection

Threshold value is set (e.g., 220)

Compare the minimum SSD with the threshold:

- If SSD > threshold, pixel is a corner
- If SSD ≤ threshold, pixel is not a corner

In our Example:

- SSD5 = 227, Threshold = 220
- Since 227 > 220, the pixel is a CORNER

Moravec Corner Detection

Conclusion

Key Takeaways:

- SSD helps measure pixel neighborhood differences
- Minimum SSD value is selected for comparison
- Thresholding determines if a pixel is a corner

Applications:

- Feature detection in computer vision
- Edge and corner detection for object tracking
- Image registration & recognition

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Perception

Feature Extraction Based on Range Data

Feature Extraction Based on Range Data (Laser, Ultrasonic)

- Most features extracted from ranging sensors in modern applications are geometric primitives, such as line segments or circles.
- Line extraction is particularly common due to its simplicity and relevance in tasks like robot localization and map building.
- The main challenges in line extraction in unknown environments include determining the number of lines present, assigning points to their respective lines, and estimating line model parameters.
- Six popular line extraction algorithms for 2D range scans will be discussed, selected based on performance and popularity in mobile robotics and computer vision.

Feature Extraction Based on Range Data (Laser, Ultrasonic)

Line Fitting

- Geometric feature fitting involves comparing and matching sensor data with a predefined description or template of the expected feature.
- Typically, the system is overdetermined, meaning there are more sensor measurements than feature parameters to be estimated.
- Due to measurement errors, there is no perfectly consistent solution, leading to an optimization problem.
- Methods such as least-squares estimation can be used to fit the feature that minimizes the discrepancy with all sensor measurements used.

Feature Extraction Based on Range Data (Laser, Ultrasonic)

Six line-extraction algorithms

- Geometric feature extraction, such as fitting a line, involves a more complex process than simple line fitting due to real-world scenarios.
- Range measurements acquired by a mobile robot are not always part of a single line; instead, there may be multiple lines represented in the measurement set.
- The process of dividing a set of measurements into subsets that can be interpreted individually is called **segmentation**.
- Segmentation is a crucial step in line extraction as it enables the identification and interpretation of individual lines within the measurement set.
- The following slides describes six popular line-extraction algorithms that address the segmentation process.

Algorithm 1: *Split-and-Merge*

1. Initial: set s_1 consists of N points. Put s_1 in a list L
2. Fit a line to the next set s_i in L
3. Detect point P with maximum distance d_P to the line
4. If d_P is less than a threshold, continue (go to step 2)
5. Otherwise, split s_i at P into s_{i1} and s_{i2}, replace s_i in L by s_{i1} and s_{i2}, continue (go to 2)
6. When all sets (segments) in L have been checked, merge collinear segments.

Feature Extraction Based on Range Data

Algorithm 2: *Line-Regression*

1. Initialize sliding window size N_f
2. Fit a line to every N_f consecutive points
3. Compute a line fidelity array. Each element of the array contains the sum of Mahalanobis distances between every three adjacent windows
4. Construct line segments by scanning the fidelity array for consecutive elements having values less than a threshold
5. Merge overlapped line segments and recompute line parameters for each segment

Feature Extraction Based on Range Data

Algorithm 3: *Incremental*

1. Start by the first 2 points, construct a line
2. Add the next point to the current line model
3. Recompute the line parameters by line fitting
4. If it satisfies the line condition, continue (go to step 2)
5. Otherwise, put back the last point, recompute the line parameters, return the line
6. Continue with the next two points, go to step 2

Algorithm 4: *RANSAC*

1. Initial: let A be a set of N points
2. **repeat**
3. Randomly select a sample of 2 points from A
4. Fit a line through the 2 points
5. Compute the distances of all other points to this line
6. Construct the inlier set (i.e. count the number of points with distance to the line $< d$)
7. Store these inliers
8. **until** Maximum number of iterations k reached
9. The set with the maximum number of inliers is chosen as a solution to the problem

Algorithm 5: *Hough Transform*

1. Initial: let A be a set of N points
2. Initialize the accumulator array by setting all elements to 0
3. Construct values for the array
4. Choose the element with max. votes V_{max}
5. If V_{max} is less than a threshold, terminate
6. Otherwise, determine the inliers
7. Fit a line through the inliers and store the line
8. Remove the inliers from the set, go to step 2

Feature Extraction Based on Range Data

Algorithm 6: *Expectation Maximization*

1. Initial: let A be a set of N points
2. **repeat**
3. Randomly generate parameters for a line
4. Initialize weights for remaining points
5. **repeat**
6. *E-Step*: Compute the weights of the points from the line model
7. *M-Step*: Recompute the line model parameters
8. **until** Maximum number of steps reached or convergence
9. **until** Maximum number of trials reached or found a line
10. If found, store the line, remove the inliers, go to step 2
11 Otherwise, terminate

Feature Extraction Based on Range Data

Table 4.4 Comparison of algorithms for line extraction from 2D laser data.

	Complexity	Speed [Hz]	False positives	Precision
Split-and-Merge	$N \cdot \log N$	1500	10%	+++
Incremental	$S \cdot N^2$	600	6%	+++
Line-Regression	$N \cdot N_f$	400	10%	+++
RANSAC	$S \cdot N \cdot N_{Trials}$	30	30%	++++
Hough-Transform	$S \cdot N \cdot N_C + S \cdot N_R \cdot N_C$	10	30%	++++
Expectation Maximization	$S \cdot N_1 \cdot N_2 \cdot N$	1	50%	++++

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Perception

Stereo Vision

Stereo Vision

- Stereopsis is the process in visual perception that leads to the sensation of depth from the slightly different projections of the world onto the retinas of the two eyes.
- Horizontal disparity, retinal disparity, or binocular disparity refer to the differences between the two retinal images caused by the eyes' different positions in the head.
- The brain fuses the two retinal images, using the information from the disparity, to perceive the object as a single and solid image.
- Disparity can be understood through a simple test where closing each eye alternately causes an object, like a finger, to appear to jump from left to right. The distance between the appearances of the object is the disparity.
- Computational stereopsis, or stereo vision, involves obtaining depth information from a pair of images captured by two cameras looking at the same scene from different positions.

Stereo Vision

In stereo vision we can identify two major problems:

1. The correspondence problem
2. 3D reconstruction

Figure 4.42 (Left) The STH-MDCS3 form Videre Design uses CMOS sensors, a baseline of 9 cm, an image resolution of 1280×960 at 7.5 frames per second (fps), or 640×480 at 30 fps. (Right) The Bumblebee2 from Point Grey uses CCD sensors, a baseline of 12 cm, an image resolution of 1024×768 at 20 frames per second (fps), or 640×480 at 48 fps.

Stereo Vision: Basic case

First, we consider a simplified case in which two cameras have the same orientation and are placed with their optical axes parallel, at a separation of *b* (called *baseline)* as shown in the figure.

In this figure, a point on the object is described as being at coordinate (x, y, z) with respect to the origin located in the left camera lens. The image coordinate in the left and right image are (u_l, v_l) and (u_r, v_r) respectively. From figure 4.43a and using equations (4.36) and (4.37), we can write

$$\frac{f}{z} = \frac{u_l}{x} , \tag{4.57}$$

$$\frac{f}{z} = \frac{-u_r}{b-x} , \tag{4.58}$$

from which we obtain

$$z = b\frac{f}{u_l - u_r} , \tag{4.59}$$

Stereo Vision: Basic case

where the difference in the image coordinates, $u_l - u_r$ is called disparity.

This is an important term in stereo vision, because it is only by measuring disparity that we can recover depth information.

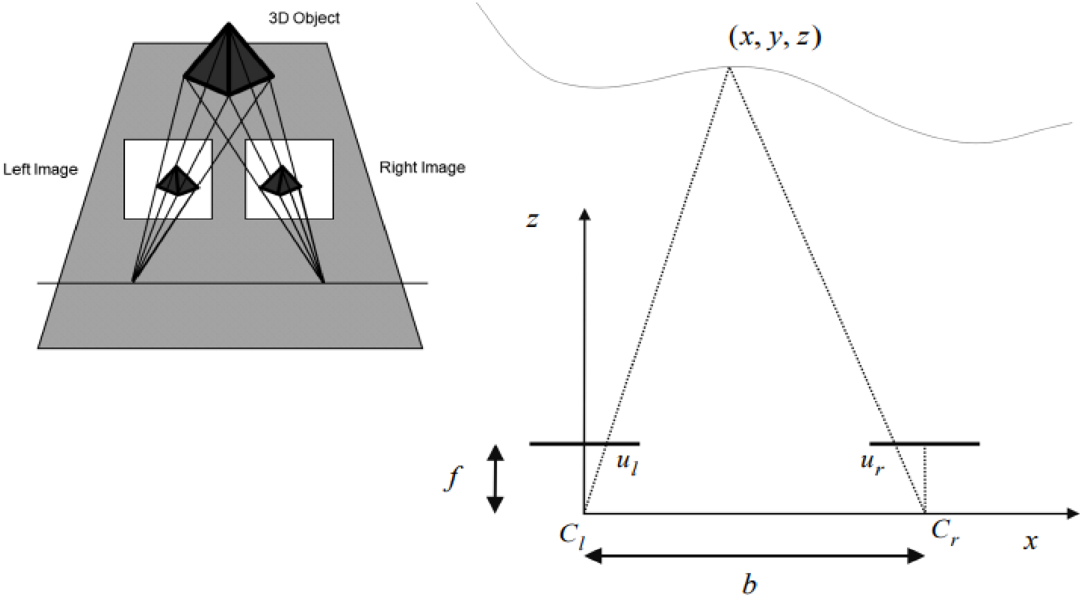

Figure 4.43
Idealized camera geometry for stereo vision. The cameras are assumed be identical (i.e., identical focal lengths and image resolution); furthermore, they are assumed to be perfectly aligned on the horizontal axis.

Stereo Vision: Basic case

Observations from this equation are as follows:

- Distance is inversely proportional to disparity. The distance to near objects can therefore be measured more accurately than that to distant objects, just as with depth from focus techniques. In general, this is acceptable for mobile robotics, because for navigation and obstacle avoidance closer objects are of greater importance.
- Disparity is proportional to *b*. For a given disparity error, the accuracy of the depth estimate increases with increasing baseline *b*.
- As *b* is increased, because the physical separation between the cameras is increased, some objects may appear in one camera but not in the other. This is due to the field of view of the cameras. Such objects by definition will not have a disparity and therefore will not be ranged.

Observations from this equation are as follows (Contd.):

- If the baseline *b* is unknown, it is possible to reconstruct the scene point only up to a scale. This is the case in structure-from-motion.
- A point in the scene visible to both cameras produces a pair of image points known as a <u>conjugate pair</u>, or <u>correspondence pair.</u>

Stereo Vision

Correspondence problem.

Using the preceding equations requires us to have identified the conjugate pair in the left and right camera images, which originates from the same scene point (figure). This fundamental challenge is called the *corresponcedence problem.*

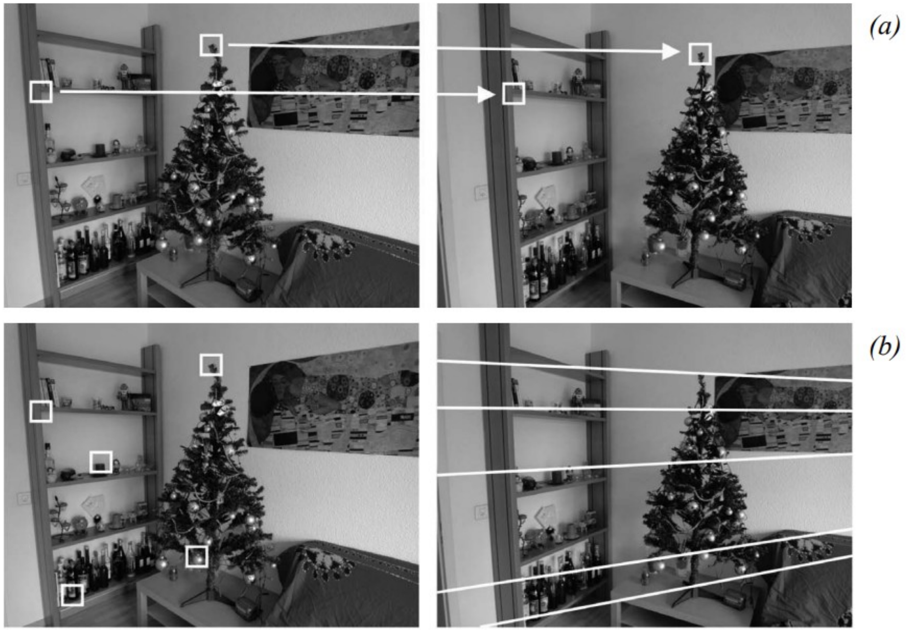

Figure 4.45 A stereo pair. Corresponding points are projections of the same scene point. Because of the epipolar constraint, conjugate points can be searched along the epipolar lines. This heavily reduces the computational cost of the correspondence search: from a two-dimensional search it becomes a one-dimensional search problem.

Stereo Vision: Correspondence problem

Using an opportune image similarity metric, a given point in the first image can be paired with one point in the second image.

The problem of false correspondences makes the correspondence search challenging. False correspondences occur when a point is paired to another that is not its real conjugate.

This is because the assumption of image similarity does not hold very well, for instance if the part of the scene to be paired appears under different illumination or geometric conditions.

Stereo Vision : Correspondence problem

Other problems that make the correspondence search difficult are:

- Occlusions
- Photometric distortion
- Projective distortion

Stereo Vision: Correspondence problem

Some constraints can be exploited for improving the correspondence search, which are :

- Similarity constraint
- Continuity constraint
- Unicity
- Monotonic order
- Epipolar constraint
- Area-based
- Feature-based

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